RSAC 2026 Conference

POWER OF COMMUNITY.

Beyond Credentials: Victim Profiling in the Stealer Malware Economy

Andréanne Bergeron

Researcher
Flare

Olivier Bilodeau

Principal Cybersecurity Researcher
Flare

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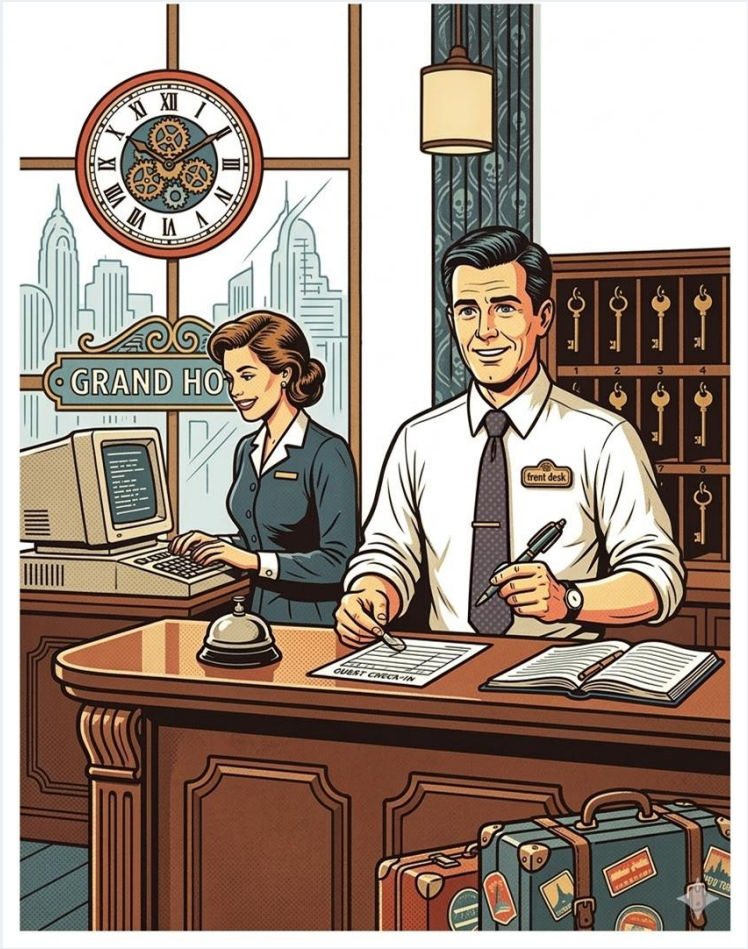
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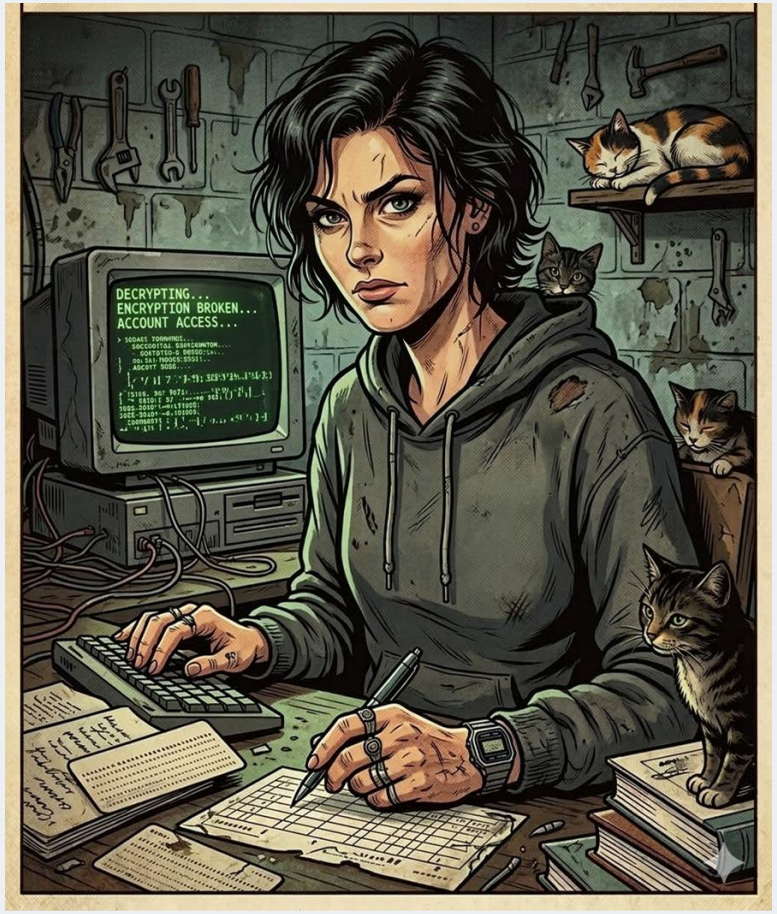
What Do These Have in Common?



A Director in a 1000+ Employees US Telco

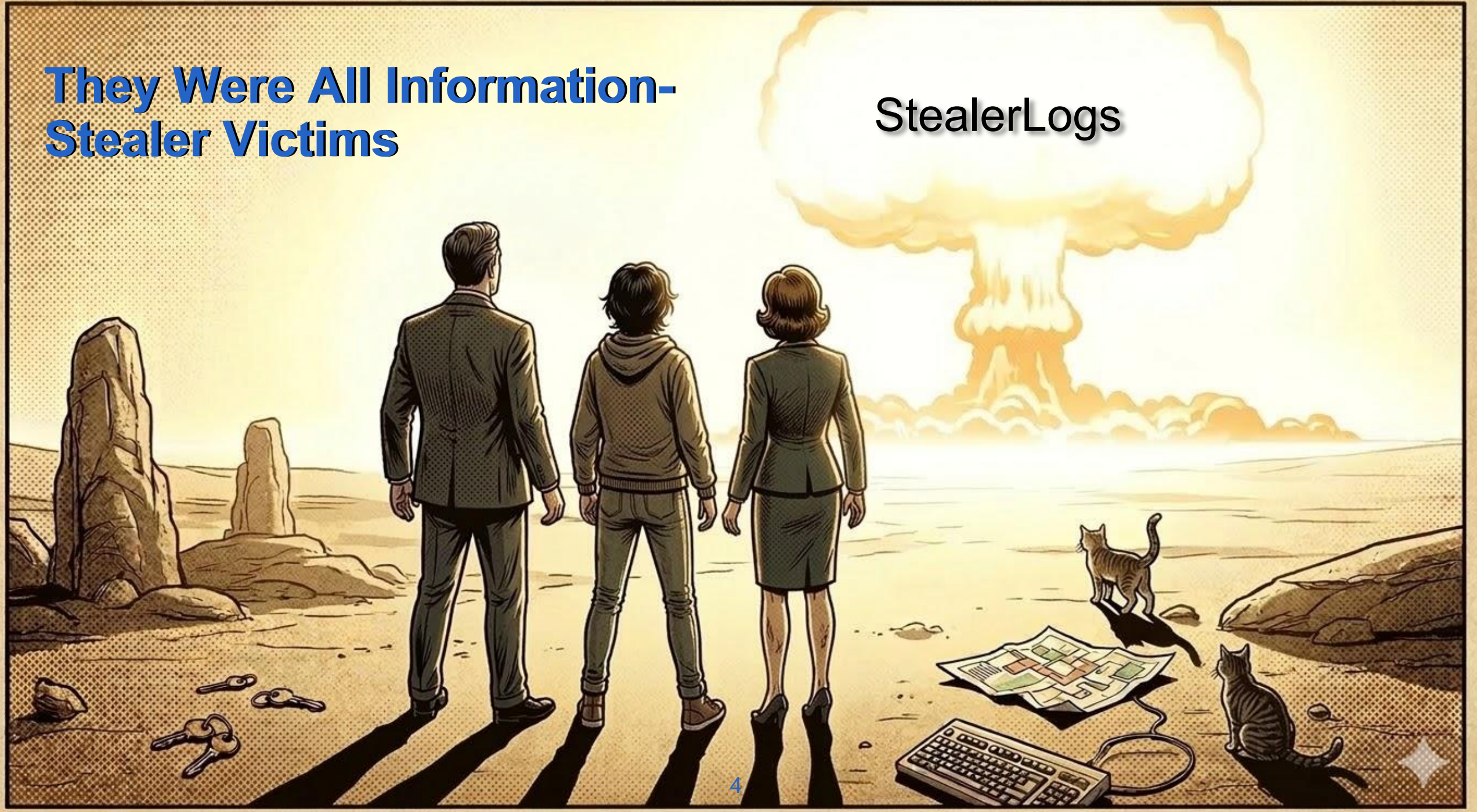


A Hotel Reception Desk



A Threat Actor

They Were All Information-Stealer Victims



Forbes

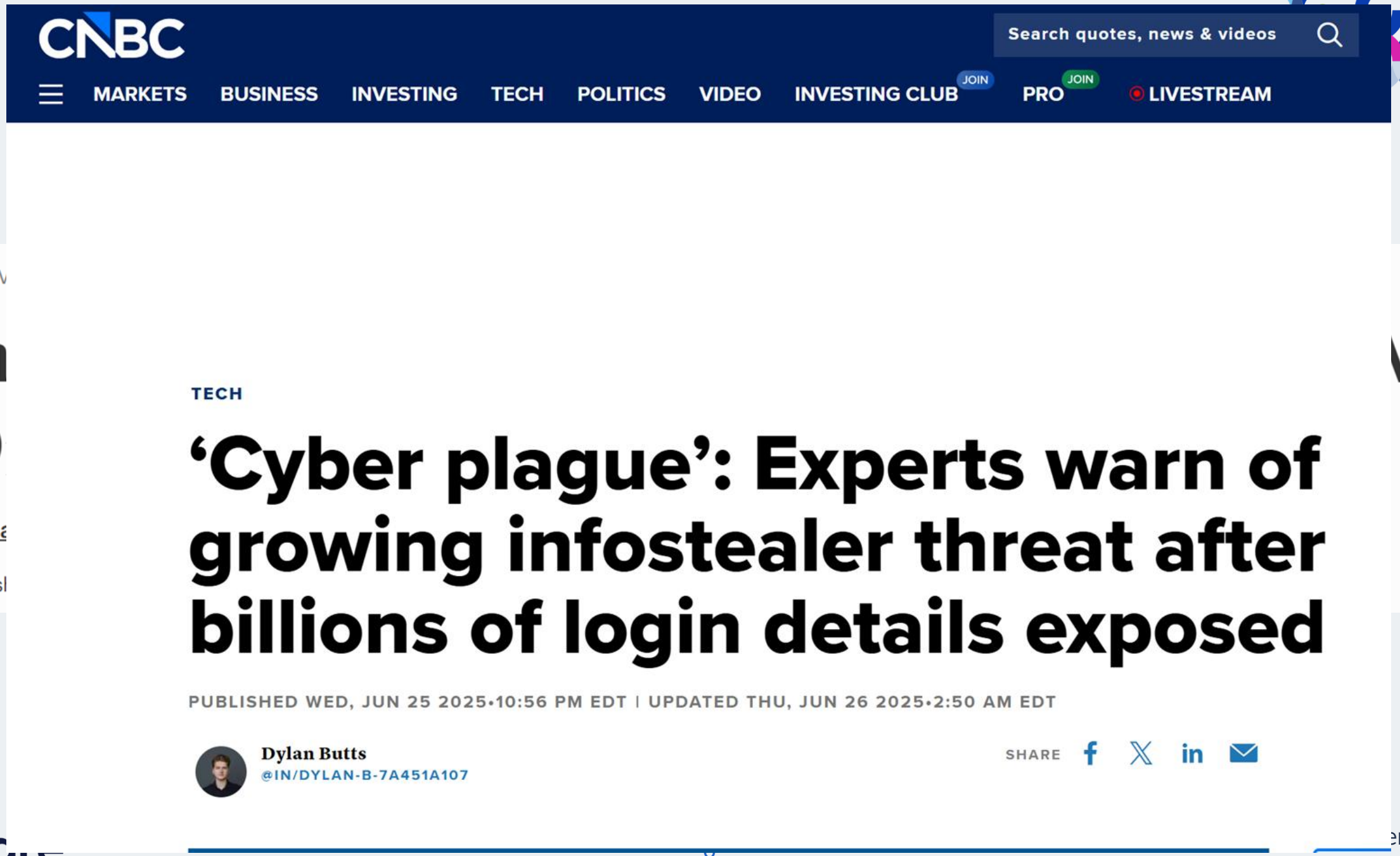
INNOVATION > CYBERSECURITY

Infostealer Infects 1 Million Windows Devices Via Discord, Dropbox, GitHub

By Davey Winder, Senior Contributor. ⓘ Davey Winder is a veteran cybersecur...

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‘Cyber plague’: Experts warn of growing infostealer threat after billions of login details exposed

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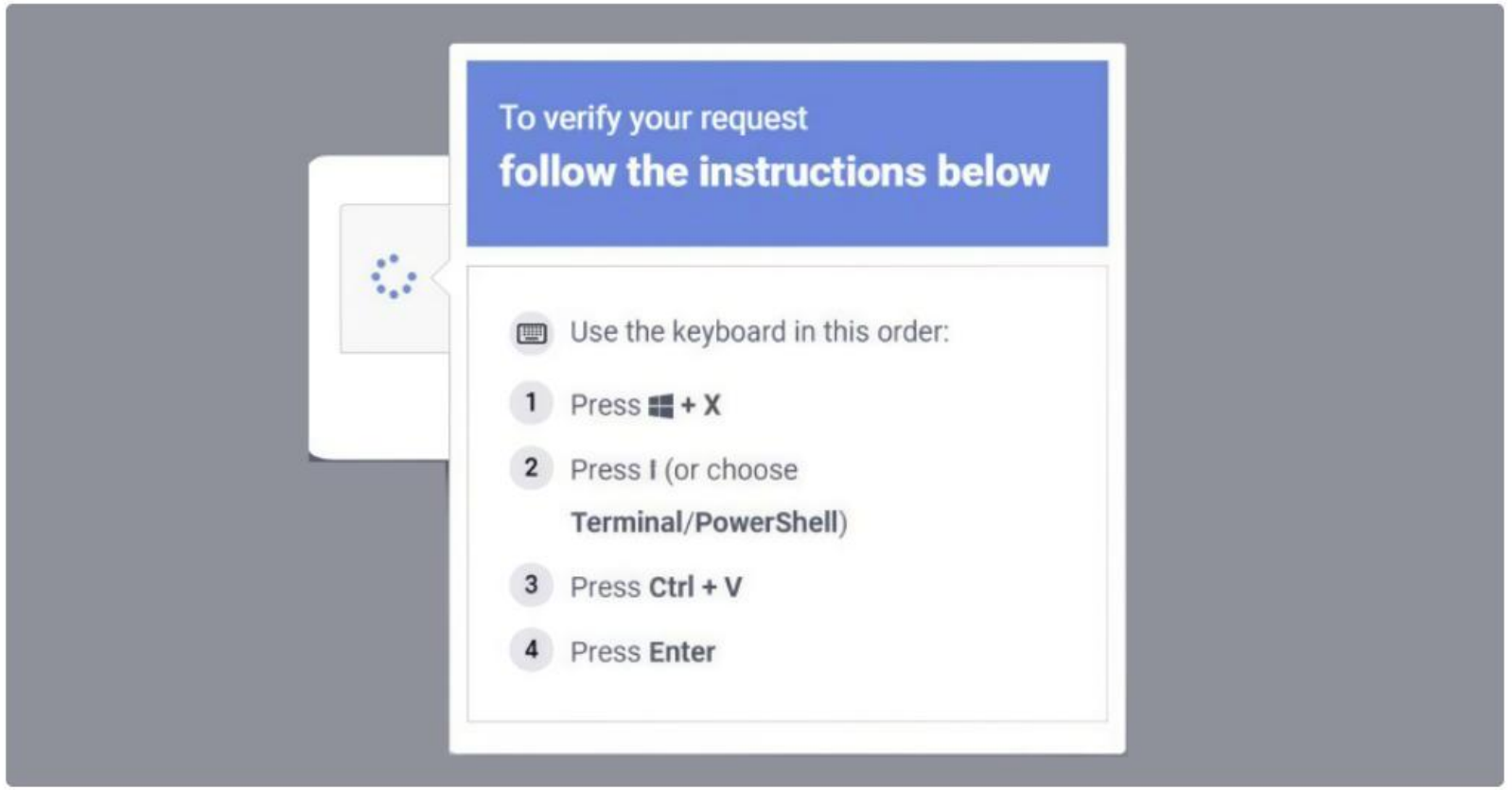
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Microsoft Reveals ClickFix Campaign Using Windows Terminal to Deploy Lumma Stealer

Ravie Lakshmanan Mar 06, 2026



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Hundreds Targeted in New Atomic macOS Stealer Campaign

Between June and August, over 300 entities were targeted with the Atomic macOS Stealer via malvertising.



By Ionut Arghire | August 22, 2025 (4:04 AM ET)





flare
ence

NEWS

Europol and Microsoft disrupt world's largest infostealer Lumma

Public-private partnerships are a cornerstone of Europol's work in the digital age

Agenda

- What are Information Stealer Malware and StealerLogs
- Our Objective: Victim Profiling
- Approach and Dataset
- Infection Vectors Identified
- Profiles and Awareness Insights
- What Have We Learned?

team_df = pandas.read_csv("team.csv")

Andréanne Bergeron

- Ph.D. in Criminology
- Security Researcher at Flare
- Affiliated Professor, Université de Montréal
- VP Conference at Northsec
- Presented at Black Hat USA, DEFCON, Shmoocon

Olivier Bilodeau

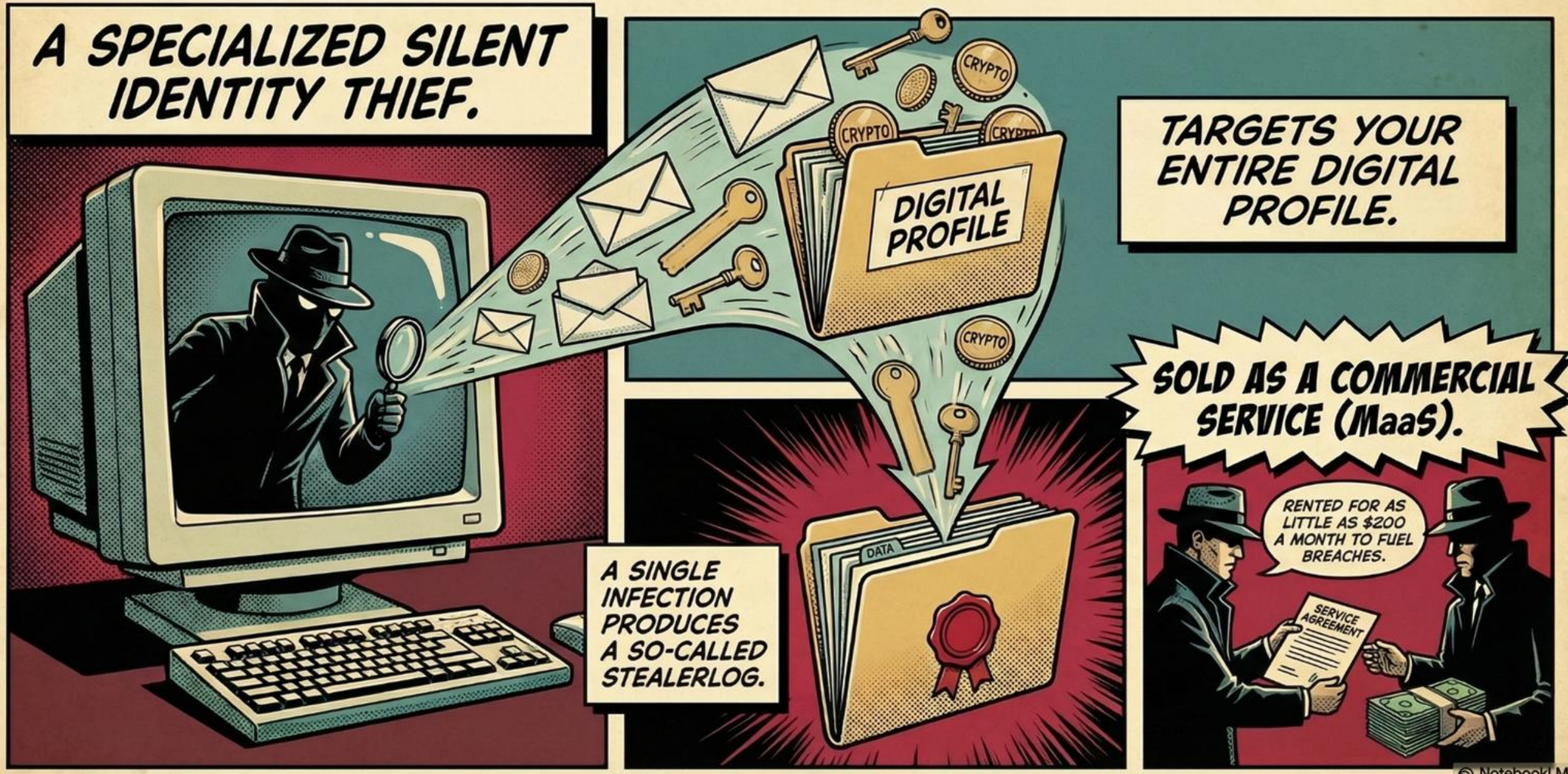
- 15 years Cybersecurity Industry XP
- Principal Cybersecurity Researcher at Flare
- Founder MontréHack
- NorthSec Vice-President
- Presented: RSAC, DEFCON, BlackHat, SecTor, Botconf, CERT-EU, AtlSecCon

Special shoutout: Estelle Ruellan



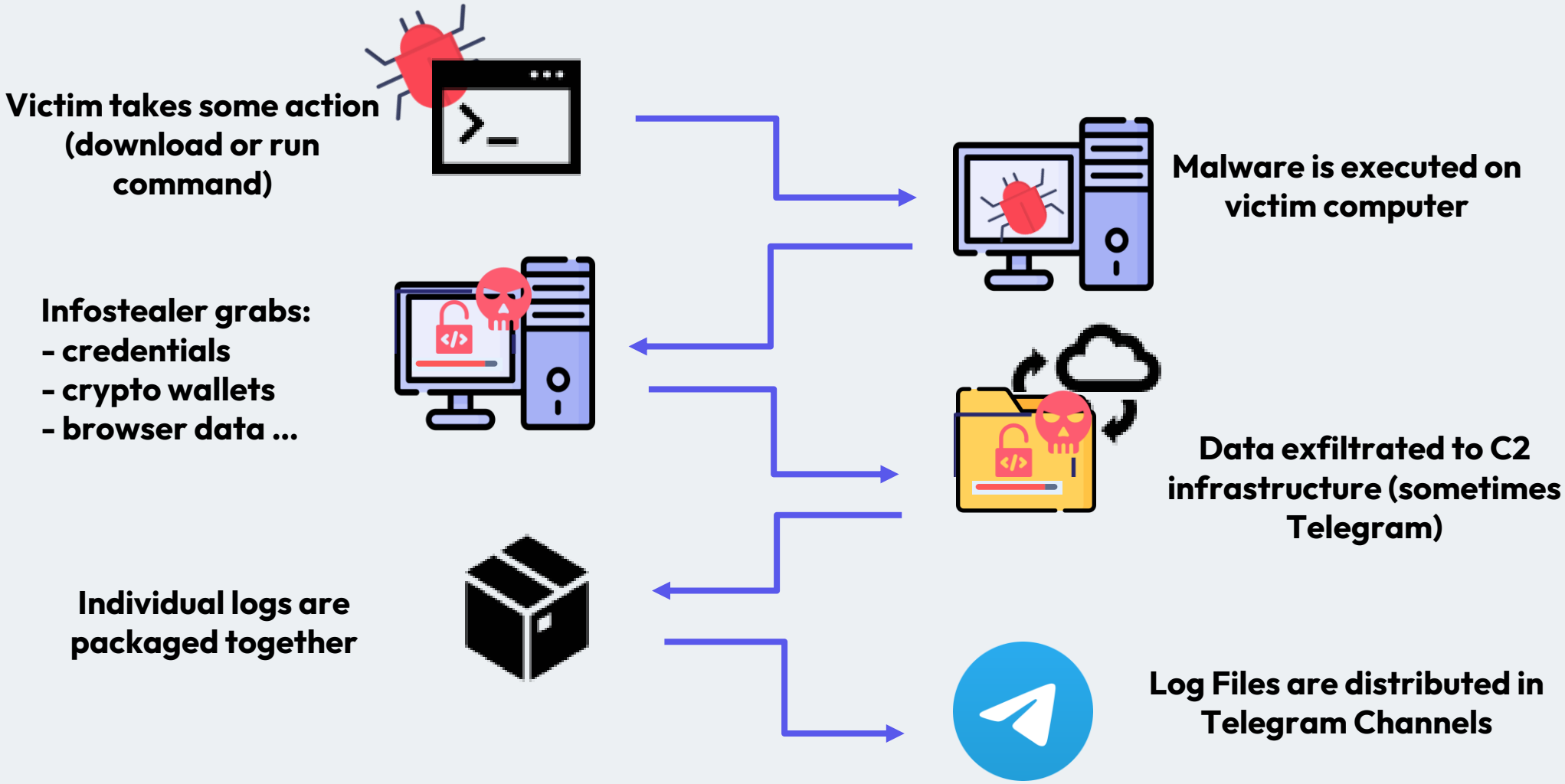

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THE SILENT SHADOW: WHAT IS AN INFOSTEALER?



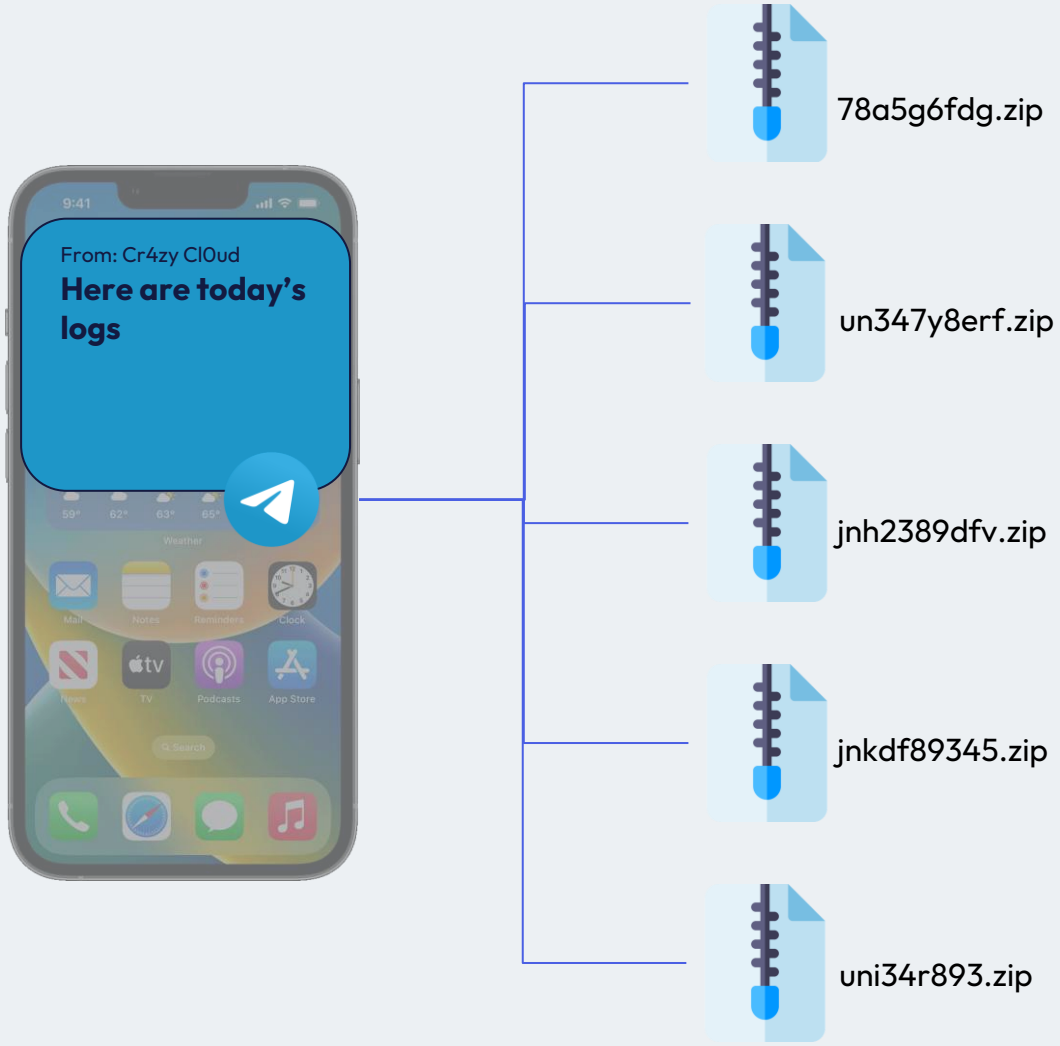
NotebookLM

The Malware you may have never heard of

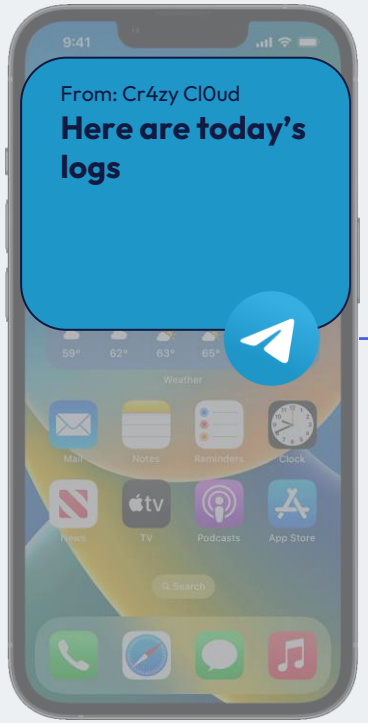


No administrative rights required and no persistence!

Stealer Log Structure



Stealer Log Structure



78a5g6fdg.zip

un347y8erf.zip

jnh2389dfv.zip

jnkdf89345.zip

uni34r893.zip

SystemInfo.txt

Passwords.txt

Screenshot.jpg

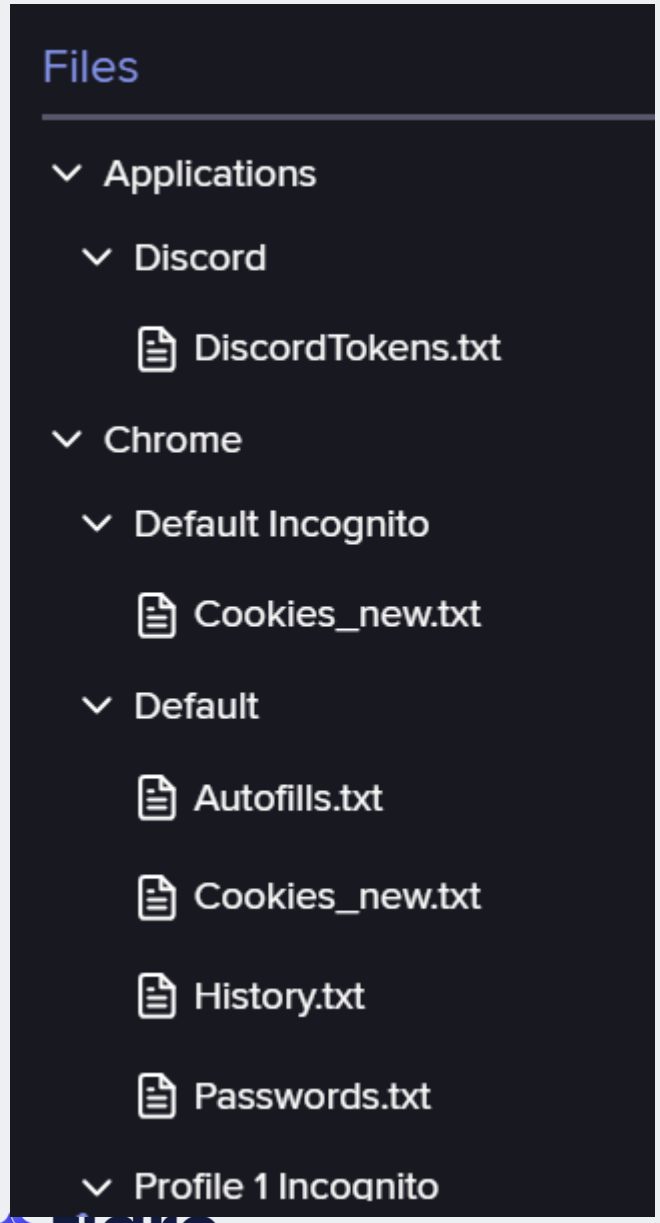
cookies

chrome_profile_1.txt

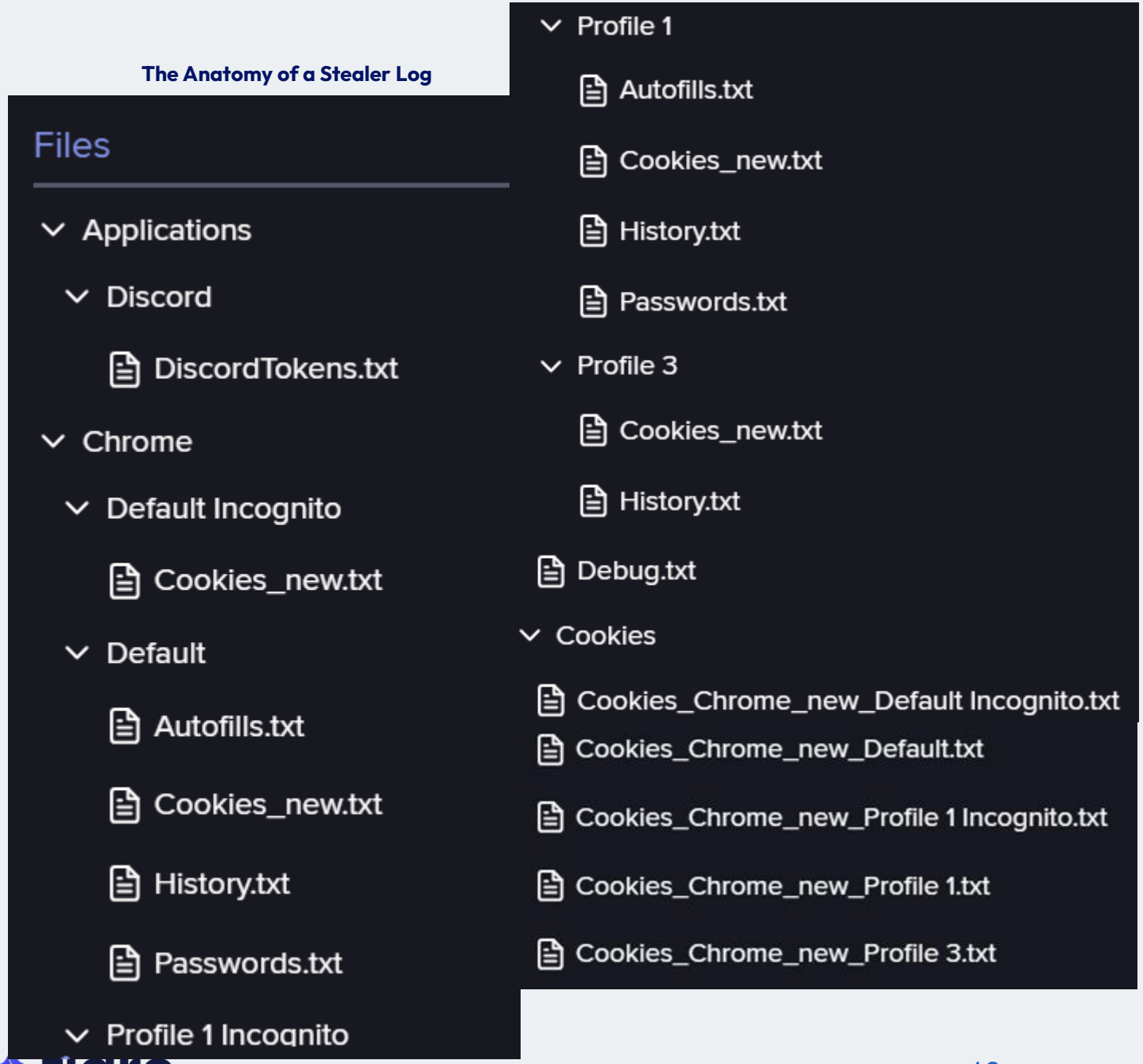
Opera_profile_2.txt

files

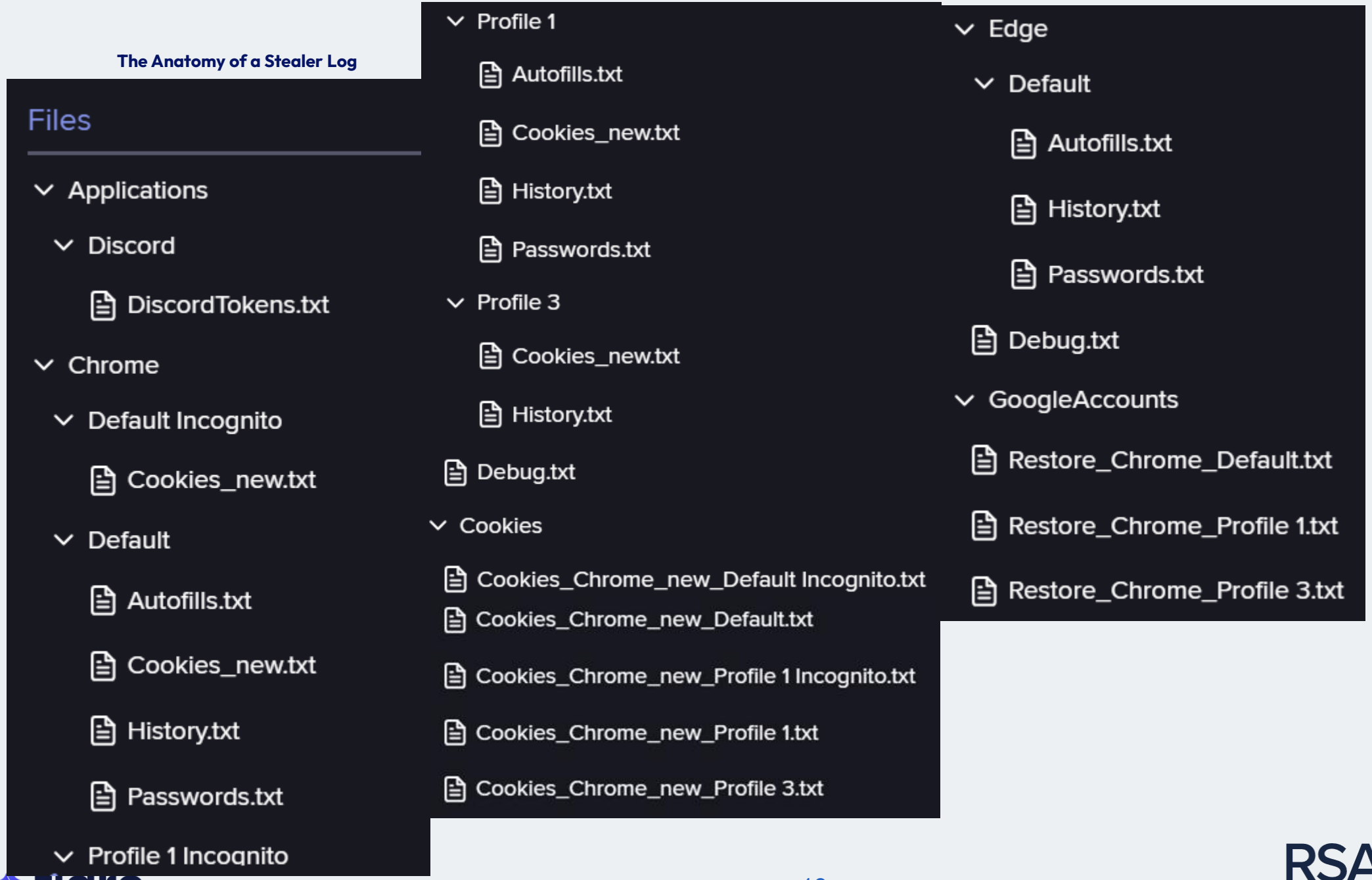
The Anatomy of a Stealer Log



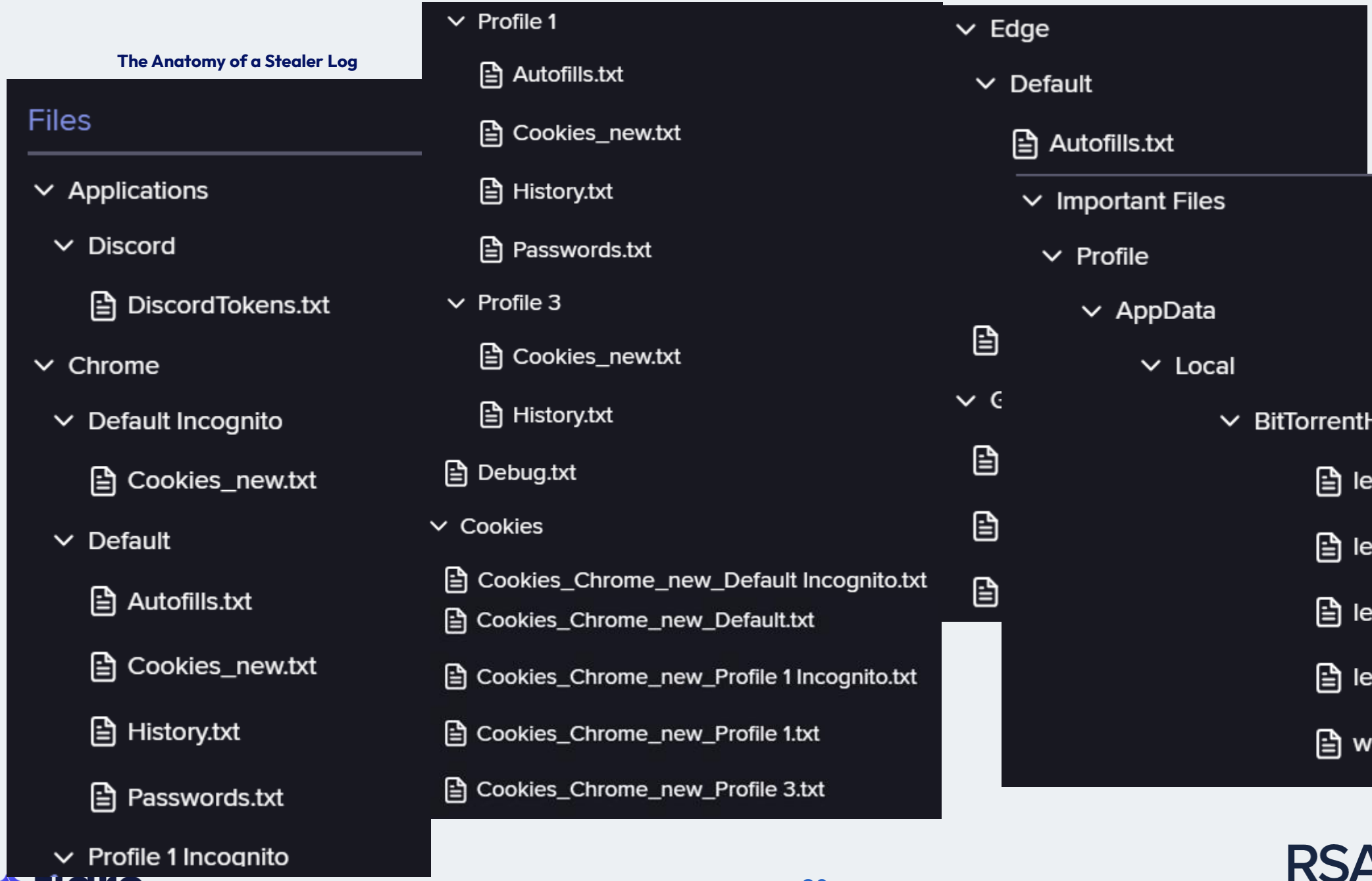
The Anatomy of a Stealer Log



The Anatomy of a Stealer Log

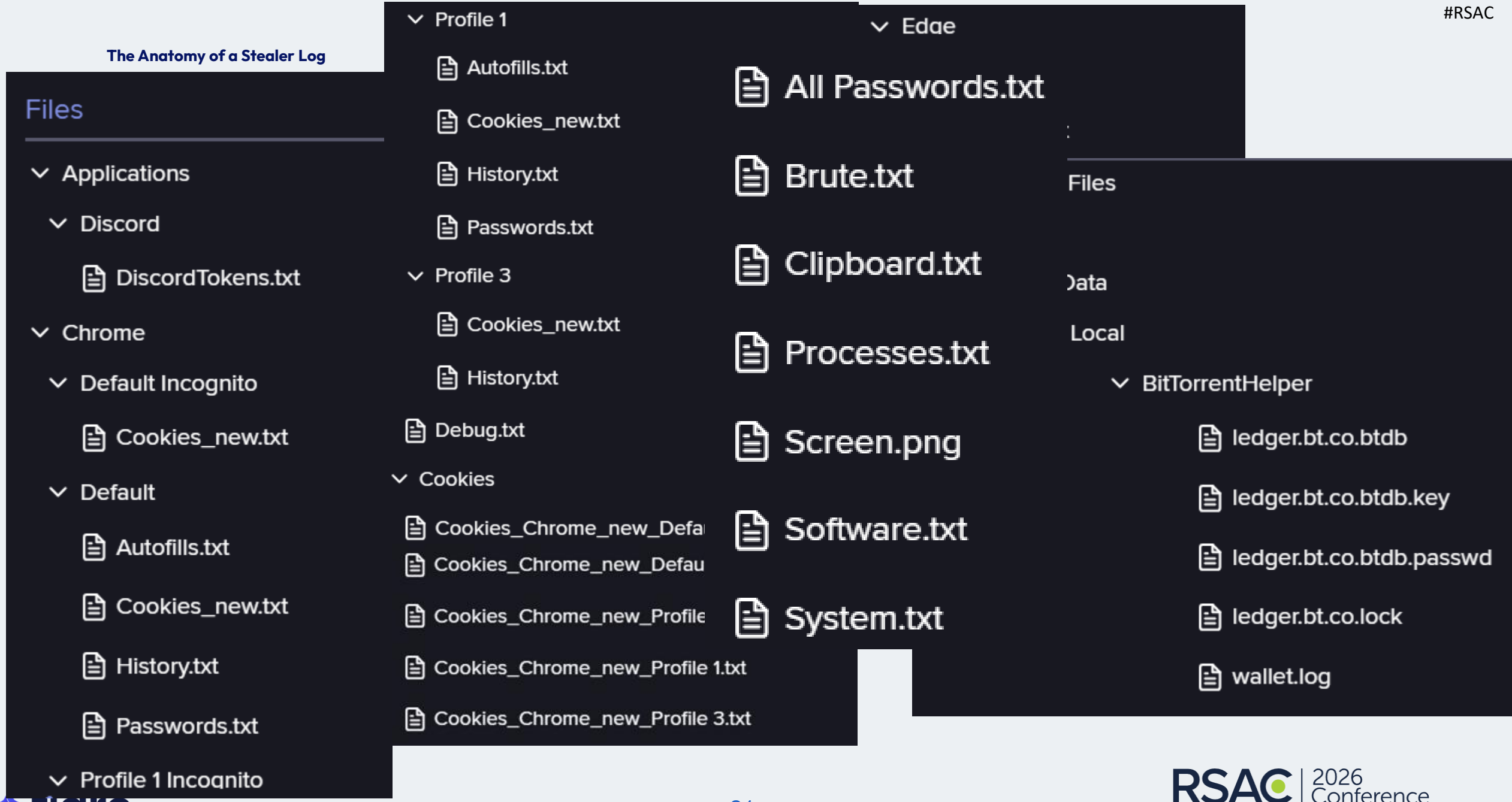


The Anatomy of a Stealer Log



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The Anatomy of a Stealer Log



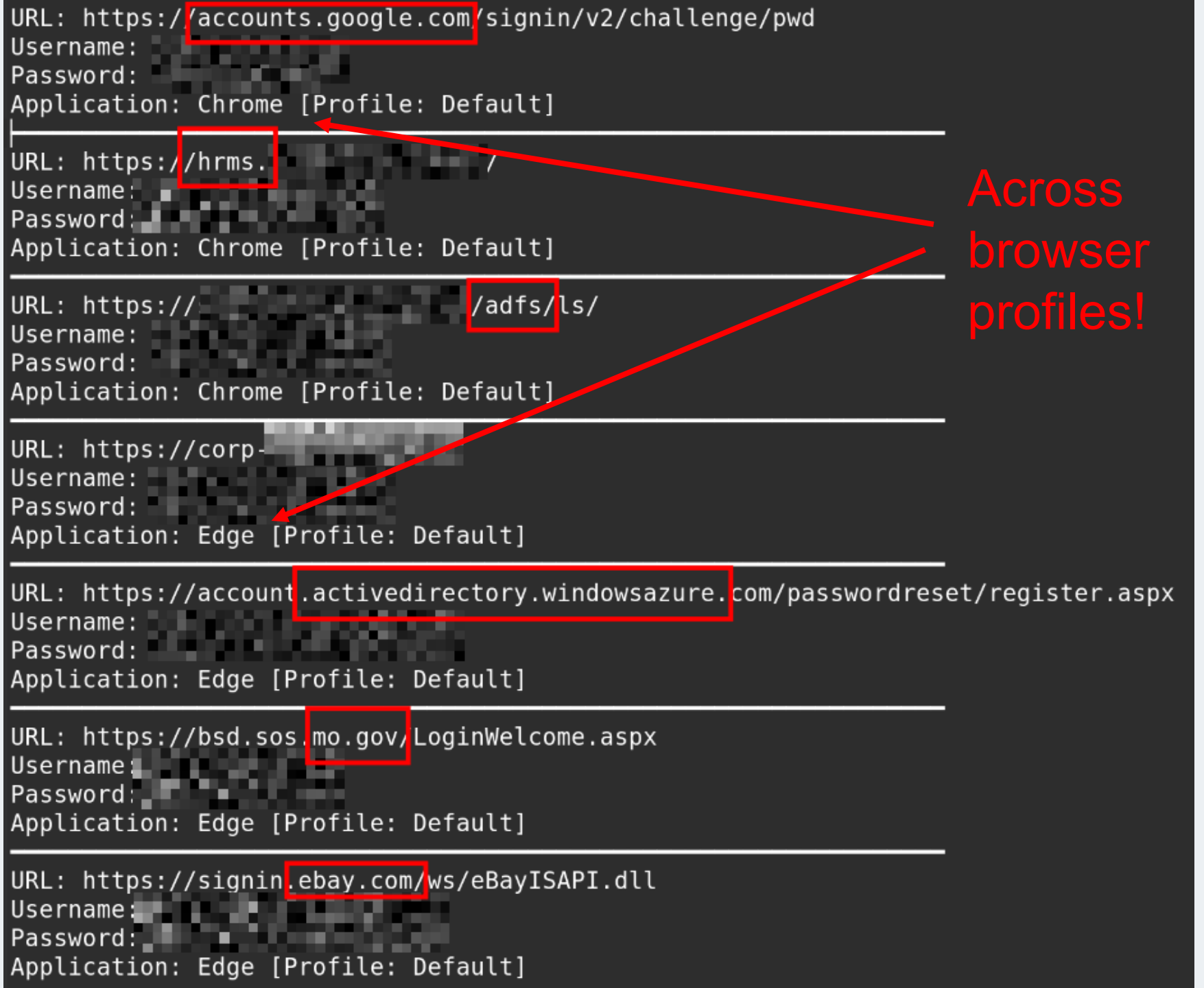
The Anatomy of a Stealerlog

passwords.txt

```
URL: https://accounts.google.com/signin/v2/challenge/pwd
Username:
Password:
Application: Chrome [Profile: Default]
--------------------------------------------------------------------------------
URL: https://hrms.                                    /
Username:
Password:
Application: Chrome [Profile: Default]
--------------------------------------------------------------------------------
URL: https://                                  /adfs/ls/
Username:
Password:
Application: Chrome [Profile: Default]
--------------------------------------------------------------------------------
URL: https://corp-
Username:
Password:
Application: Edge [Profile: Default]
--------------------------------------------------------------------------------
URL: https://account.activedirectory.windowsazure.com/passwordreset/register.aspx
Username:
Password:
Application: Edge [Profile: Default]
--------------------------------------------------------------------------------
URL: https://bsd.sos.mo.gov/LoginWelcome.aspx
Username:
Password:
Application: Edge [Profile: Default]
--------------------------------------------------------------------------------
URL: https://signin.ebay.com/ws/eBayISAPI.dll
Username:
Password:
Application: Edge [Profile: Default]
```

flare

passwords.txt



flare

History.txt

```
URL: https://translate.google.co.id/m?sl=auto&tl=id&q=This%20app%20may%20cause%20damage%20to%20your%20device.
%20Sensitive%20personal%20data%20may%20also%20be%20at%20risk.%20More%20info&hl=id
TITLE: Google Terjemahan
TIME: 18.06.2025 19:32:28

URL: https://allstrongservices.ca/resetter-printer-epson-l3150/
TITLE: Privacy error
TIME: 18.06.2025 19:33:30

URL: https://tdsafe.org/ybnV4Y?s=allstrongservices.ca&mv=v
TITLE: Privacy error
TIME: 18.06.2025 19:33:30

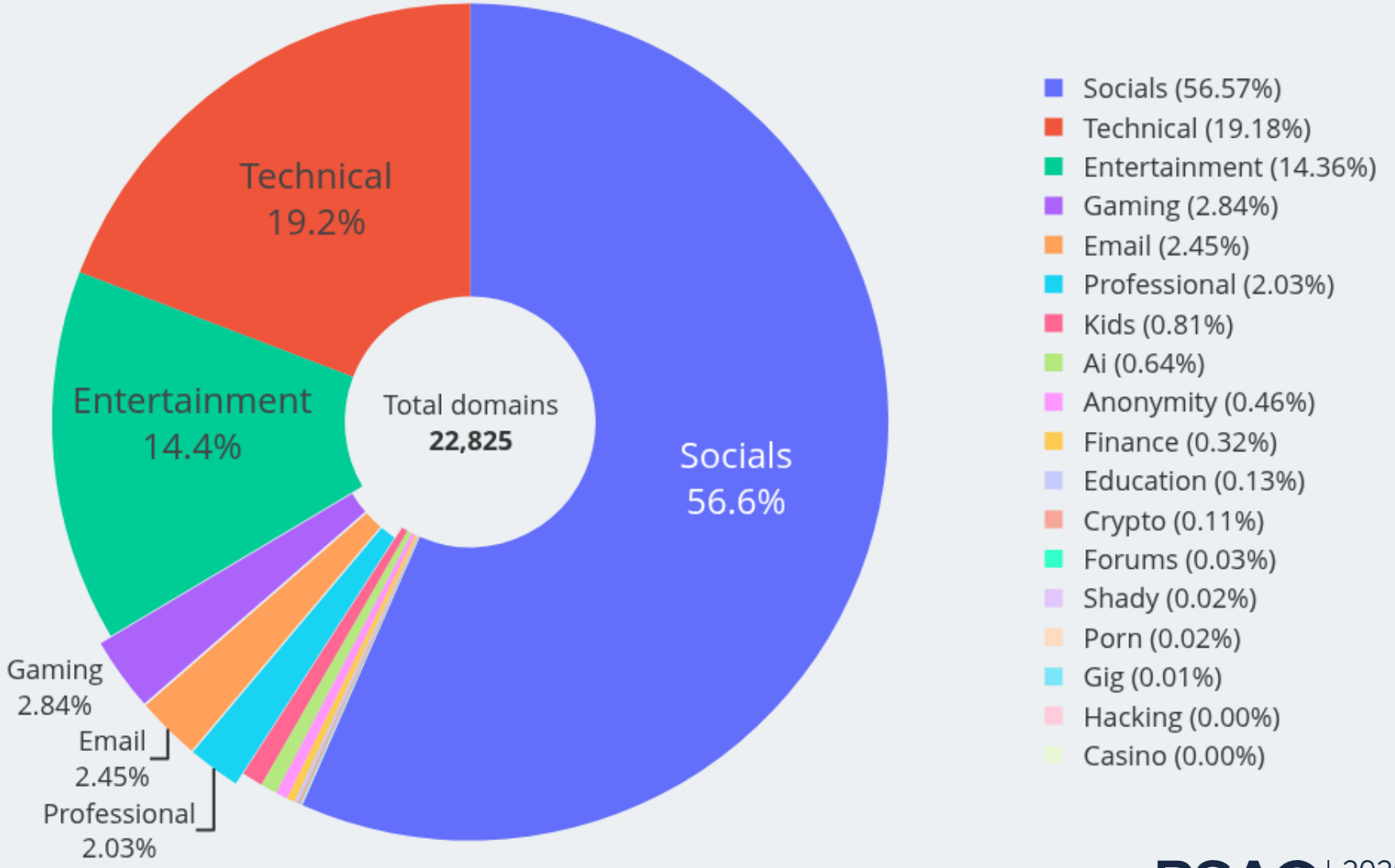
URL: https://1wprde.com/casino/list/4?p=j03k
TITLE: Privacy error
TIME: 18.06.2025 19:33:30

URL: https://www.google.com/search?q=resetter+epson+l3150&sca_esv=7b640742644021d7&ei=2ehSaLb9C-aP4-
EPyr6XgQg&start=90&sa=N&sstk=Ac65TH6lv3GijgBZ0ali_MsL1eDNpKdH_g-
vYqz7NBCKU5vgxNyMjbpFYAqze2eNPJYjJPdgd4HrFzIADIVIYsq3I8w2qQjPUYUDWw&ved=2ahUKEwj2zfDLsfuNAxXmxzgGHUrfJYAQ8tMDegQIBRAU&biw=1366&bih=589&dpr=1
TITLE: resetter epson l3150 - Penelusuran Google
TIME: 18.06.2025 19:34:29

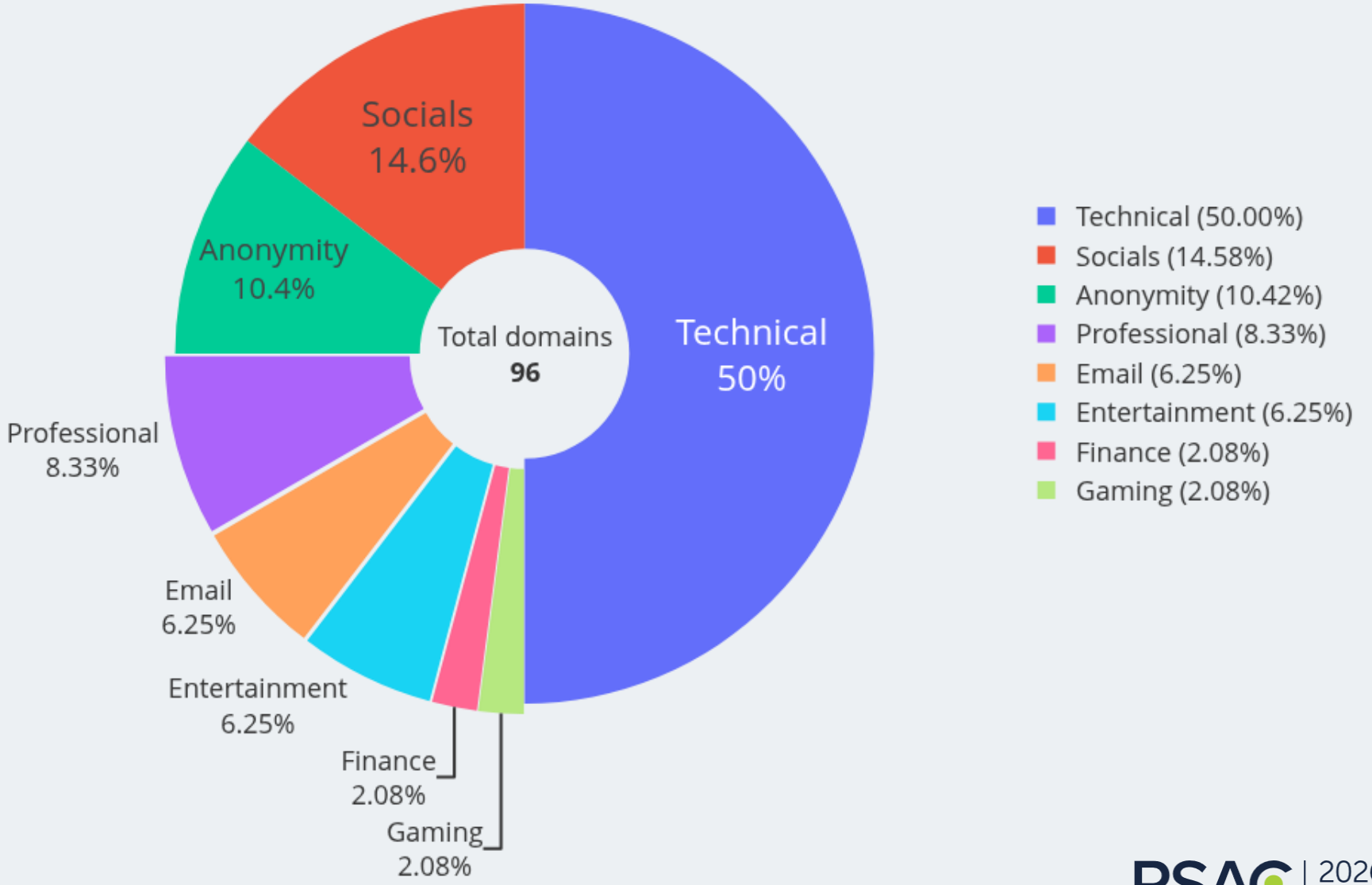
URL: https://www.google.com/search?
sca_esv=7b640742644021d7&q=Download+Resetter+Epson+L3150+Gratis+tanpa+password&sa=X&ved=2ahUKEwiTvcCfs_uNAxXOzjgGHSkbNX84WhDVAnoECCYQAQ&biw=1366&bi
TITLE: Download Resetter Epson L3150 Gratis tanpa password - Penelusuran Google
TIME: 18.06.2025 19:34:54

URL: https://tendencydigestion.icu/tracker/index.php?code=bir&p=1&q=Download%20Resetter%20Epson%20L3150%20Gratis%20(Terbaru%202025)%20-
%20Jawara%20Cloud&k=undefined
TITLE: Download Resetter Epson L3150 Gratis (Terbaru 2025) - Jawara Cloud - Free Download
TIME: 18.06.2025 19:35:33
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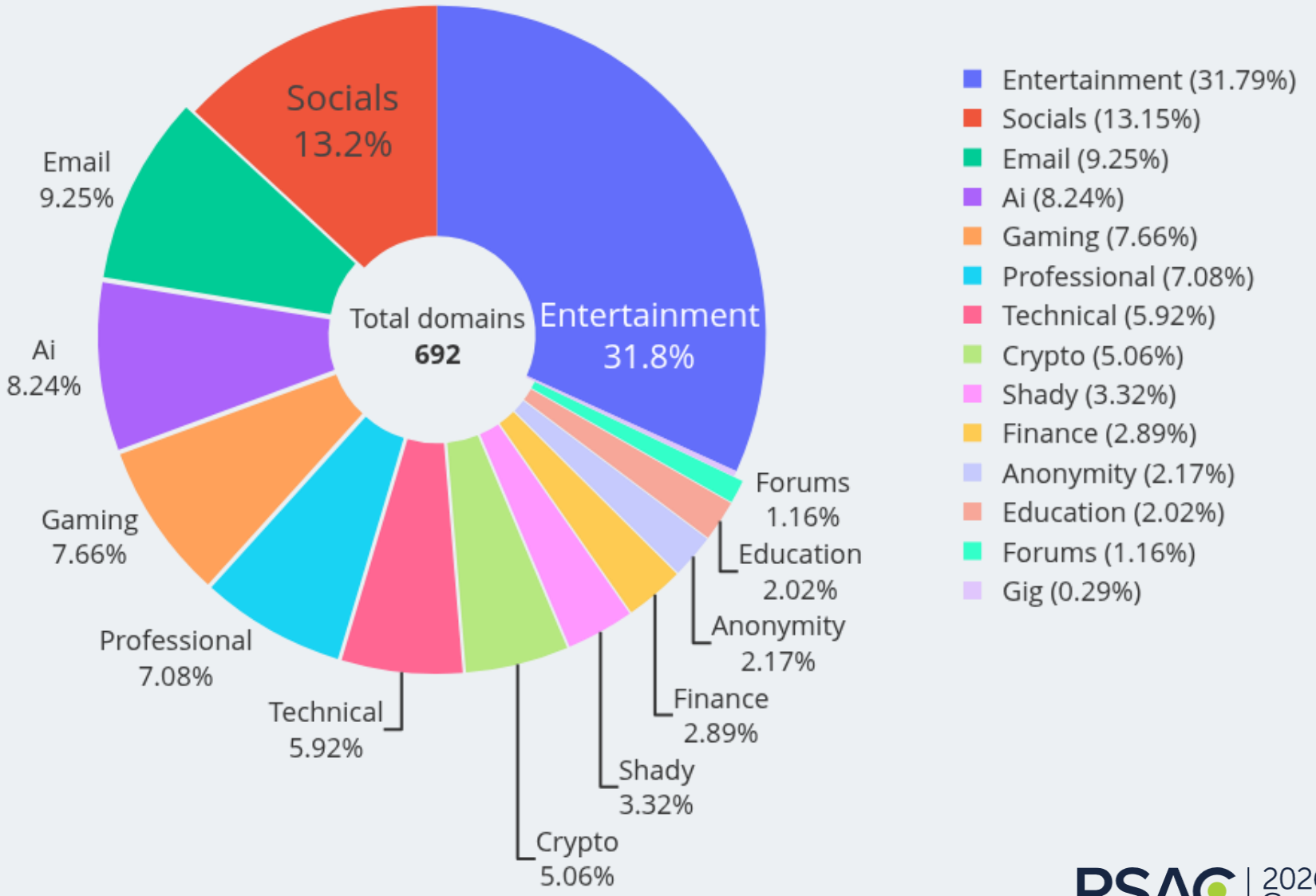
Browser History Analysis of One Victim



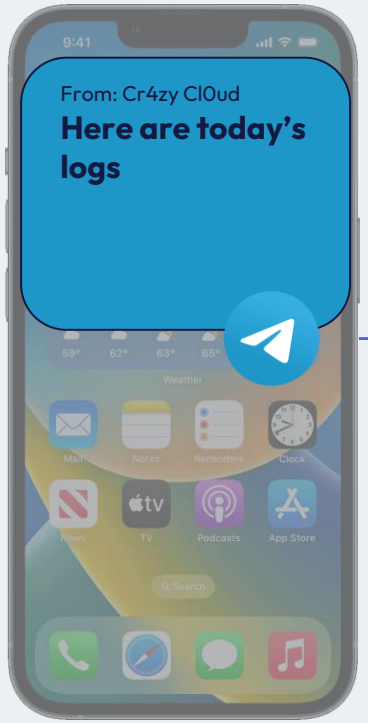
Browser Credential Analysis of One Victim



Browser Cookie Analysis of One Victim



Stealer Log Structure: The Screenshot



78a5g6fdg.zip

un347y8erf.zip

jnh2389dfv.zip

jnkdf89345.zip

uni34r893.zip

SystemInfo.txt

Passwords.txt

Screenshot.jpg

cookies

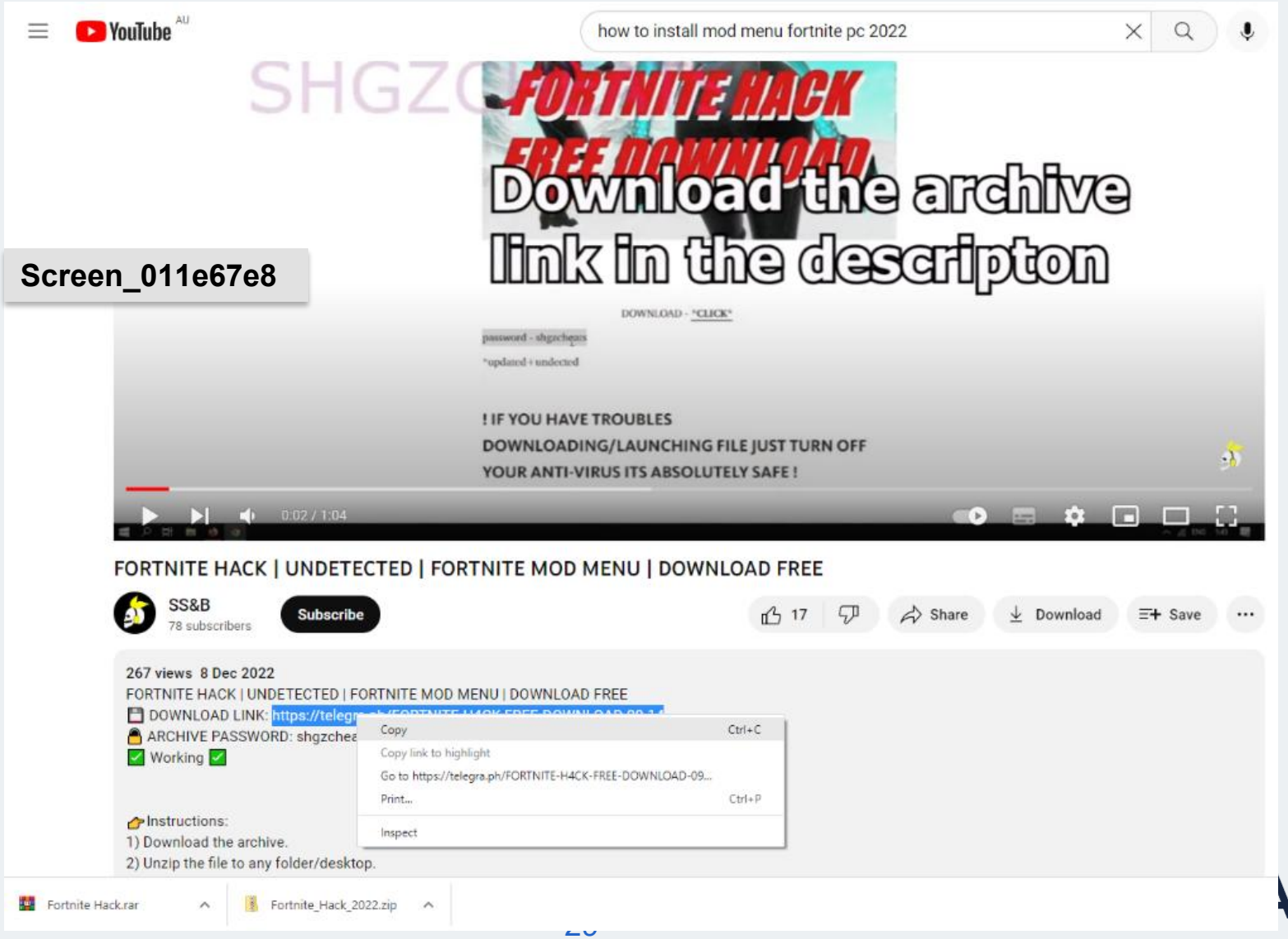
files

chrome_profile_1.txt

Opera_profile_2.txt

Stealer Log Structure: the screenshot

Screen_011e67e8



THE DIGITAL CRIME SCENE



A Visual "Crime Scene" Record

Screenshots capture the victim's desktop at the exact moment of compromise, providing a visual timestamp of the attack.

Identifying Malicious Entry Points

Images reveal critical artifacts like browser tabs, malicious URLs, and installer windows used to execute the malware.

Unmasking Social Engineering Lures

Visuals expose the specific "themes" used to trick users, such as cracked software, gaming mods, or malicious Google Ads.

Enable Rapid Campaign Takedown

Extracted URLs and file names allow defenders to identify and report active infrastructure to stop widespread campaigns.

NotebookLM

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Adapting Prevention Strategies to Victim Profiling



Awareness is Proven to Work in Fraud Prevention

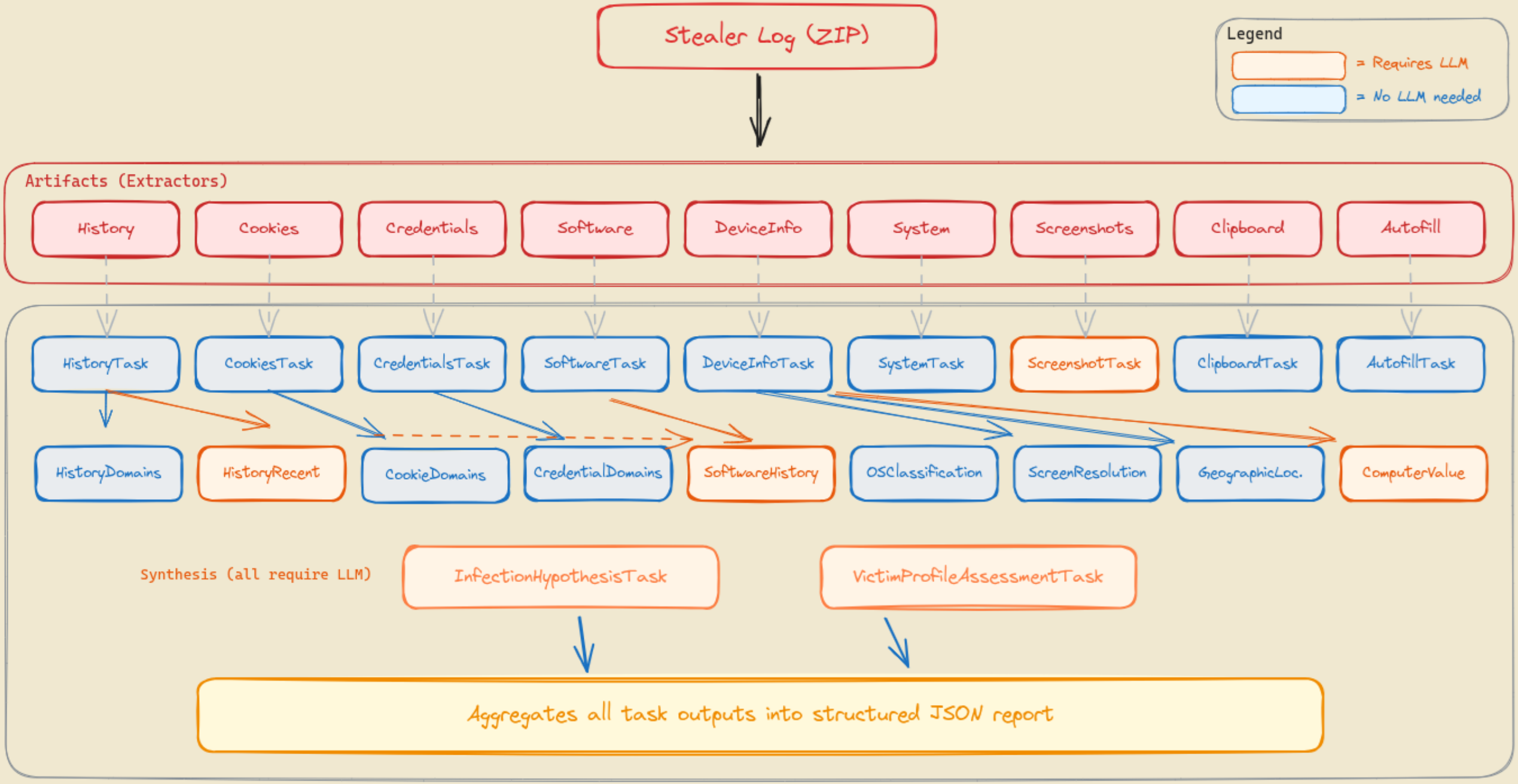
- Knowing a fraud modus operandi protects you from it
  - Relative in the hospital who “can’t talk on the phone”
  - Unknown aunt who left you an inheritance
  - Foreign prince asking for help transferring money
- The human component of cybersecurity must be actively engaged, continuously evaluated, and integrated into broader defensive strategies (National Institute of Standards and Technology)
- From passive awareness to adaptive and behaviorally informed training models

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Our Analysis Tool and Study Sample



Victim Profiling Pipeline — Analysis of a single StealerLog



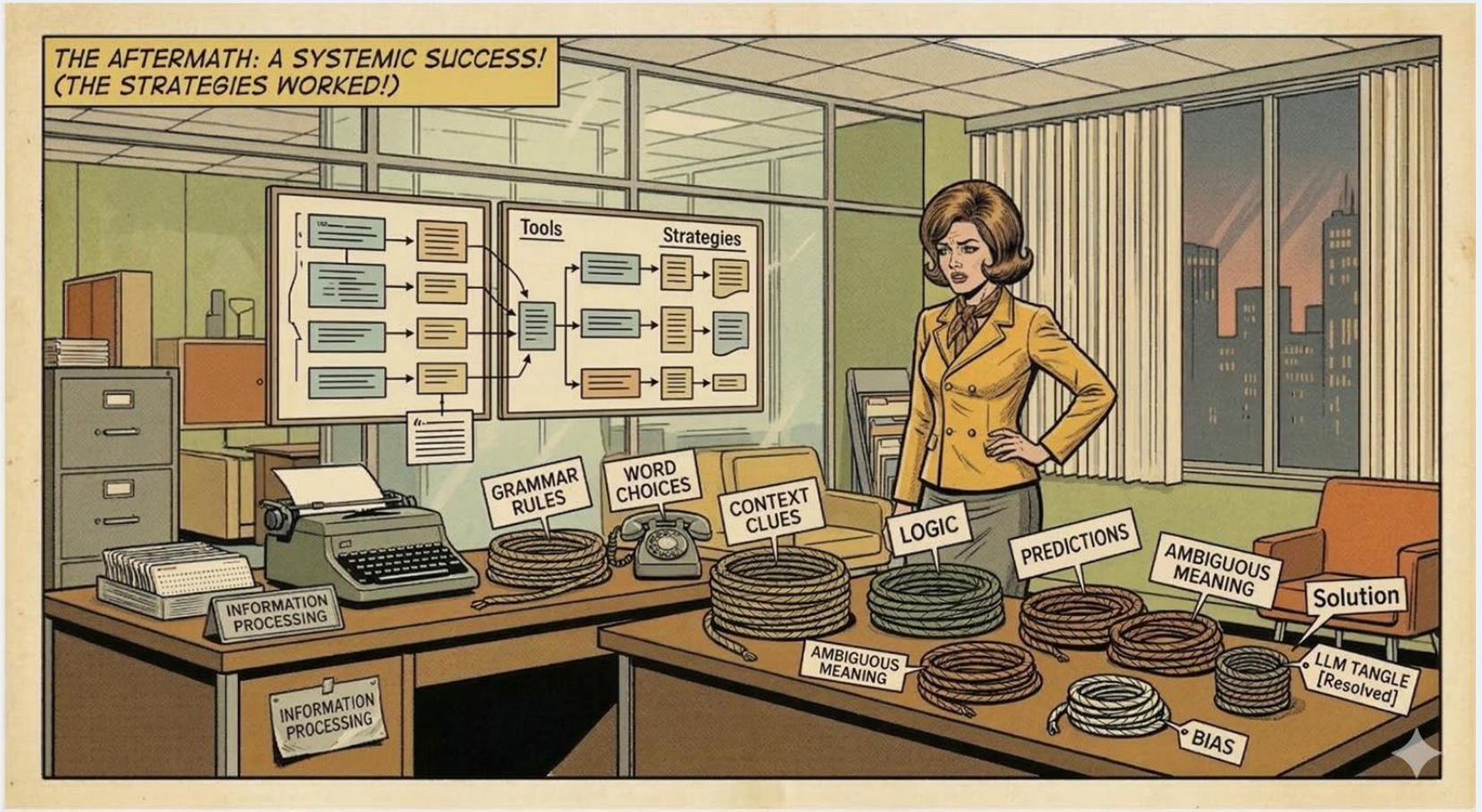
Don’t Get Fooled

- LLMs are easy to use.
- Producing useful, well-structured, relevant, and reproducible information **is much harder**



Tools and Strategies [to Sort This Out]

- Leverage structured output with Pydantic models
- Prompts must contain analyst intuition



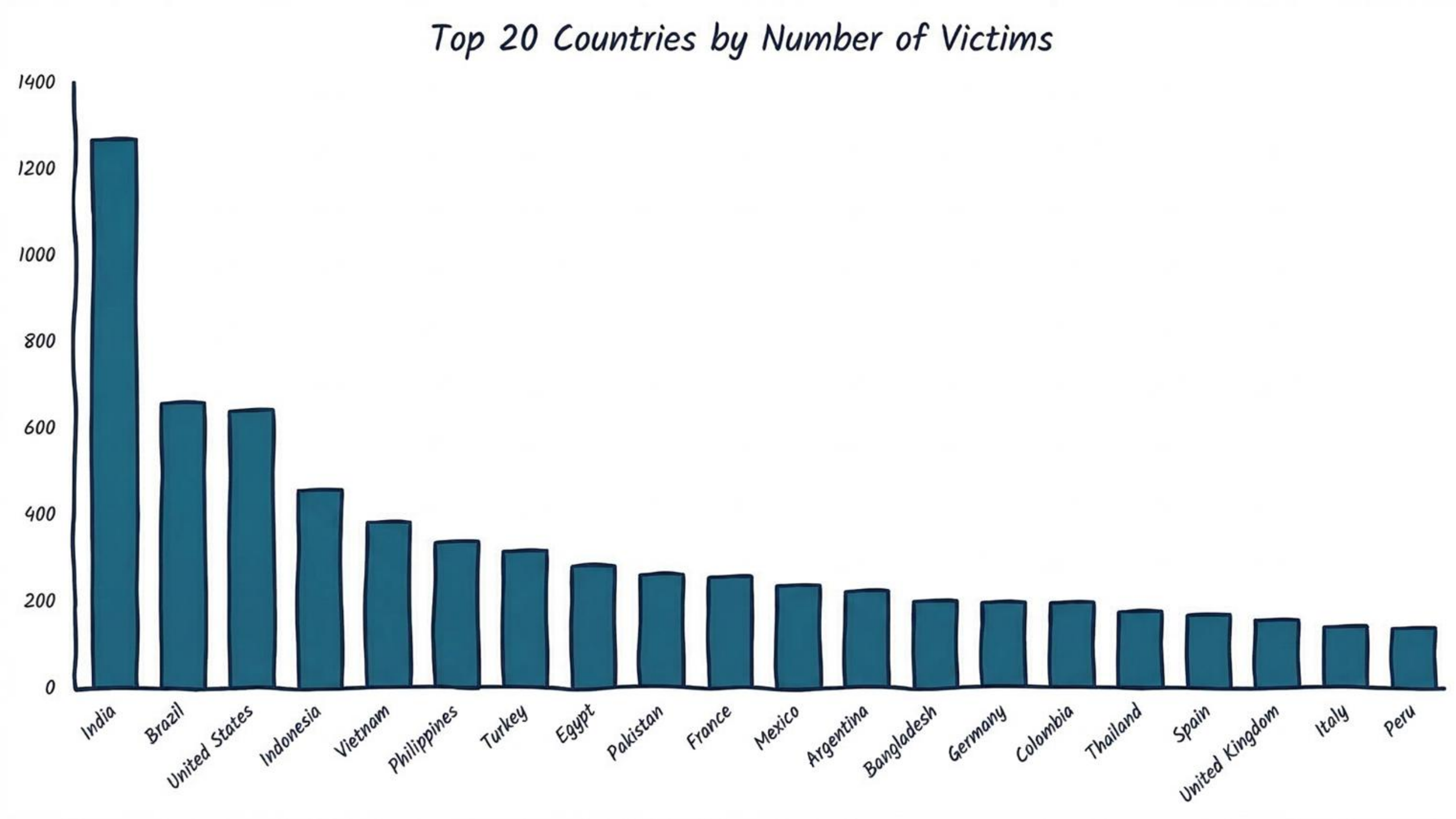
The Sample in Numbers

Random sample of approx. 30 stealer logs per day for 1 year (2025) for a total of **10 198 logs** (victims)

- ➔ 59% have financial data
- ➔ Up to 1381 personally identifiable information per victim
- ➔ **26% has company infrastructure access**
- ➔ **72% pose an organizational risk**

200k analyzed
without LLM
costs: 15$ (0.000075$/log)
storage: 360GB (1.8MB/log)

10k analyzed
with LLM
costs: 365$ (0.0365$/log)
storage: 21.6GB (2.2MB/log)
tokens: 950M (95k/log)


Top 20 Countries by Number of Victims
1400
1200
1000
800
600
400
200
0
India
Brazil
United States
Indonesia
Vietnam
Philippines
Turkey
Egypt
Pakistan
France
Mexico
Argentina
Bangladesh
Germany
Colombia
Thailand
Spain
United Kingdom
Italy
Peru

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Profiles and Awareness Posters



Important Numbers

70%

have technical tooling in their computer

+

82%

have technical skills

Technical Employee

The technical abilities lead to risky behavior: they regularly download and install software from various sources including unsigned open-source software.

Example found:

- US-based developer working for a law firm
  - Technical sophistication
    - GitHub, Keepass, FileZilla
  - Corporate and personal exposure
    - VPN, Email, Slack and Entra ID



82% of information stealer victims have technical skills

YOUR TECH SAVVINESS IS YOUR ACHILLES HEEL



FLARE-APPROVED WARNING

Scientifically verified through a year of daily infections from a dataset of more than 100M+ victims of stolen passwords.

Flare.io flare

The median information stealer victim has 83 installed software

THE MORE SOFTWARE YOU INSTALL, THE MORE DOORS YOU OPEN



FLARE-APPROVED WARNING

Scientifically verified through a year of daily infections from a dataset of more than 100M+ victims of stolen passwords.

Flare.io flare

Important Numbers

21%

of victims in the sample have more than one user per device

Shared Computers

Computers that are shared between multiple users are impacted by stealer malware: kiosks, Internet cafés, library, school labs and work shared workstations.

Example found:

- A workstation in a retail store in Malaysia
  - Access to shop management software, Point of Sale software
  - 70+ credentials and 3000+ cookies
  - Employee names and emails



Family Computer

Computer with **multiple accounts** and/or browser profiles, and autofills revealing names with identical last names.

Infection is often caused by gaming-related vectors. Leaked credentials contain a mix of entertainment, gaming and professional accounts.

Example found:

- A family in Peru
  - 22k+ history entries
  - 17k+ cookies
  - 1068 leaked credentials
  - Infected via a Valorant Skin Changer



1 in 4 information stealer infections are triggered by cracked games, cheats or mods

ROBLOX CHEATS ARE A TRAP. PROTECT YOUR KIDS, PROTECT YOUR HOME



FLARE-APPROVED WARNING

Scientifically verified through a year of daily infections from a dataset of more than 100M+ victims of stolen passwords.

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The Infection Vectors



To Each Their Own Infection Vector



Pirated Software



Services



Games and Entertainment

To Each Their Own Infection Vector



Pirated Software



Services



Games and Entertainment



The Infection Vectors: Pirated Software

Business Software

Essential Software

Creative Software

Specialised Software



The Infection Vectors: Pirated Software

Office

$$$

Office

Pirated

0$



Threat actors prey on users' willingness to bypass legitimate licensing fees at the cost of their own security

Downloading pirated software is a leading cause of information stealer infections

PIRATED SOFTWARE MIGHT COST YOU YOUR IDENTITY.



FLARE-APPROVED WARNING

Scientifically verified through a year of daily infections from a dataset of more than 100M+ victims of stolen passwords.

Flare.io ✦flare

Unlucky Executive

Professional user infected via pirated software, combined with low digital literacy. Accidental compromise.

Example found:

- A Managing Director of an outsourcing company
  - 1.2k+ unique credentials
  - Bank accounts, credit cards, HR systems, hosting providers
  - Infected via Proxy VPN installer



~50% of business software infection victims have company infrastructure access

LOOKS LIKE BUSINESS SOFTWARE. ACTS LIKE SPYWARE.

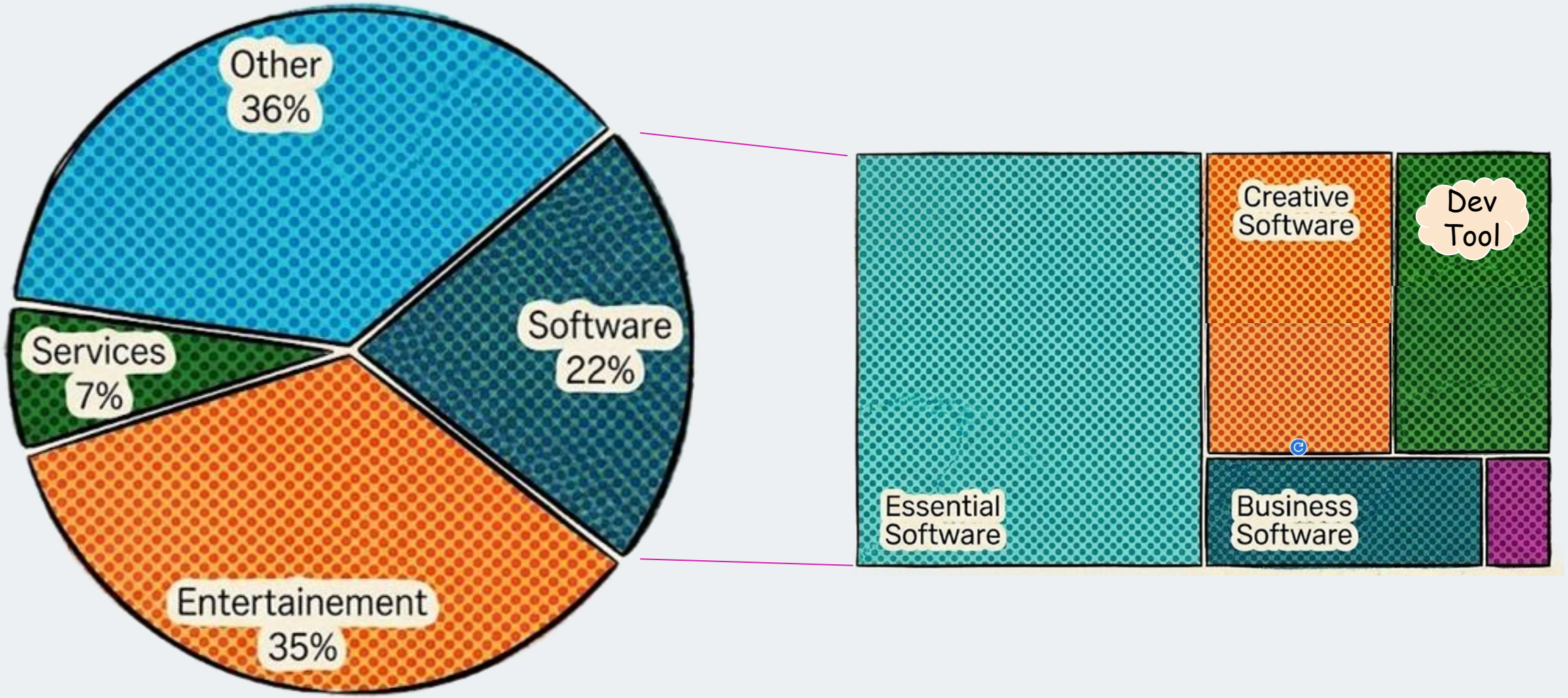


FLARE-APPROVED WARNING

Scientifically verified through a year of daily infections from a dataset of more than 100M+ victims of stolen passwords.

Flare.io flare

How people in USA get infected?



To Each Their Own Infection Vector



Pirated Software



Services



Games and Entertainment



The Infection Vectors: Services

AI

Repair Tools

Generic file-sharing

Captain Get-It-Done

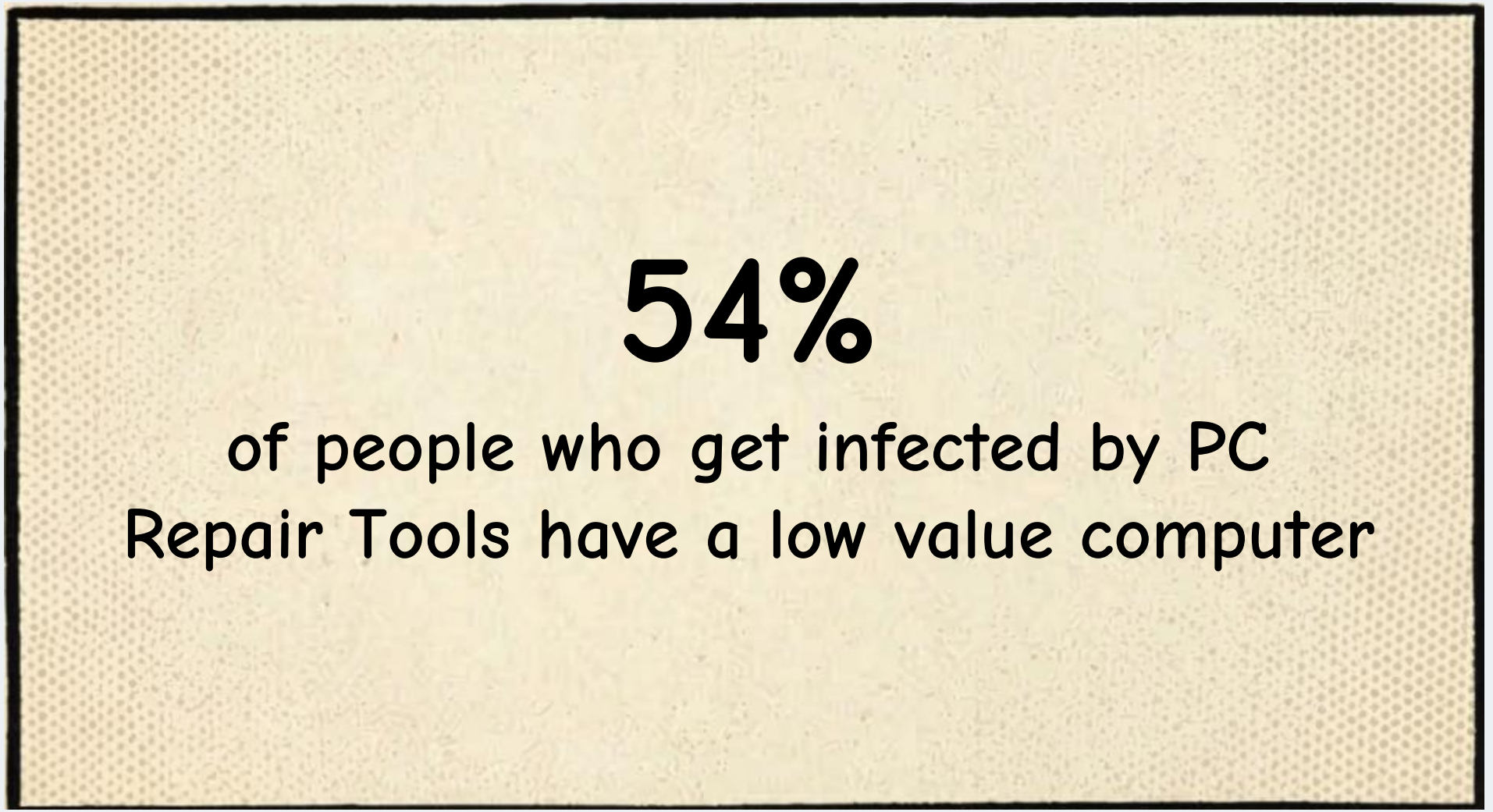
They just want to get the job done. Not tech savvy. Unfortunately, this mean downloading and executing programs to get back to work.

Example found:

- Small business owner compromised via ad redirection chain for an Epson printer adjustment utility
  - Governmental services, business email, accounting



(Not So-)Fun Fact



To Each Their Own Infection Vector



Pirated Software



Services



Games and Entertainment



The Infection Vectors: Games and Entertainment



Mainstream Classic Game

Contemporary Game

Gaming Cheat and Mods



The Infection Vectors: Games and Entertainment





The Infection Vectors: Games and Entertainment

VALORANT

FORTNITE







The Infection Vectors: Games and Entertainment

VALORANT

FORTNITE







MINECRAFT

The Teenager Gamer

Characteristically of a **young age** and having a **risk-taking behavior**, but **without** criminal intent.

Example found:

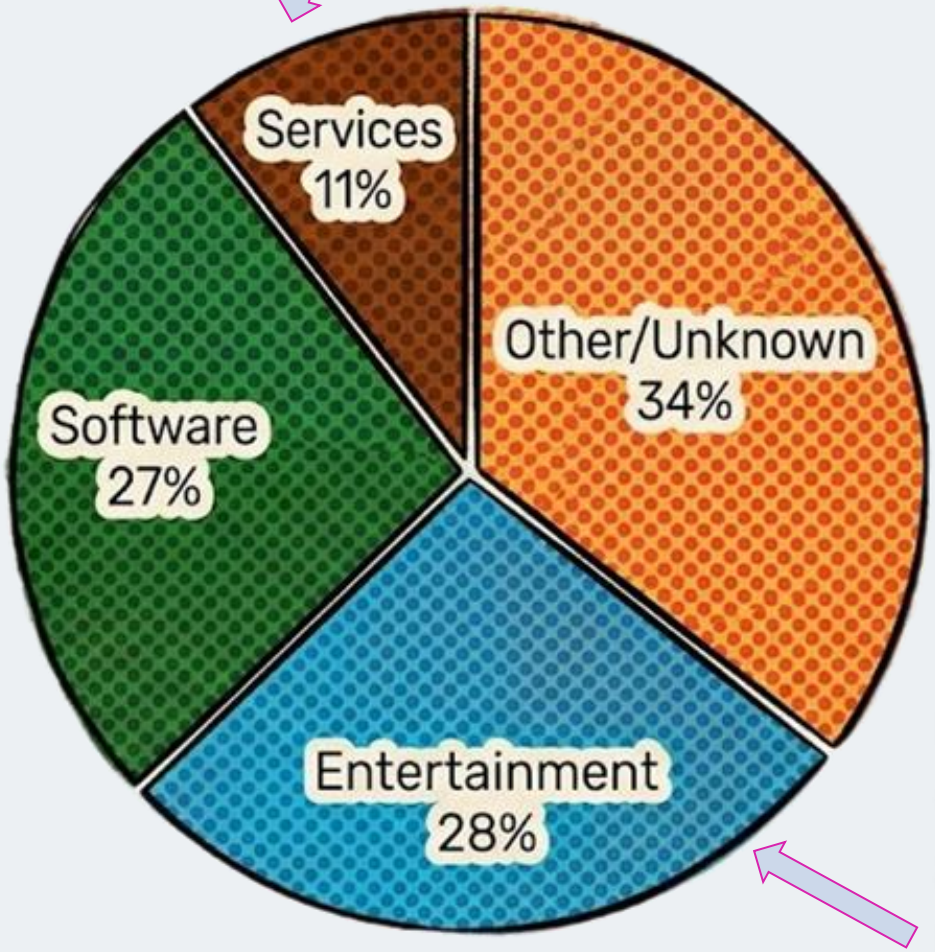
- Born in 2006 in the US, full address, name and DOB.
  - Infected via jjsploit, a Roblox cheating tool



Infection Vectors Prevalence

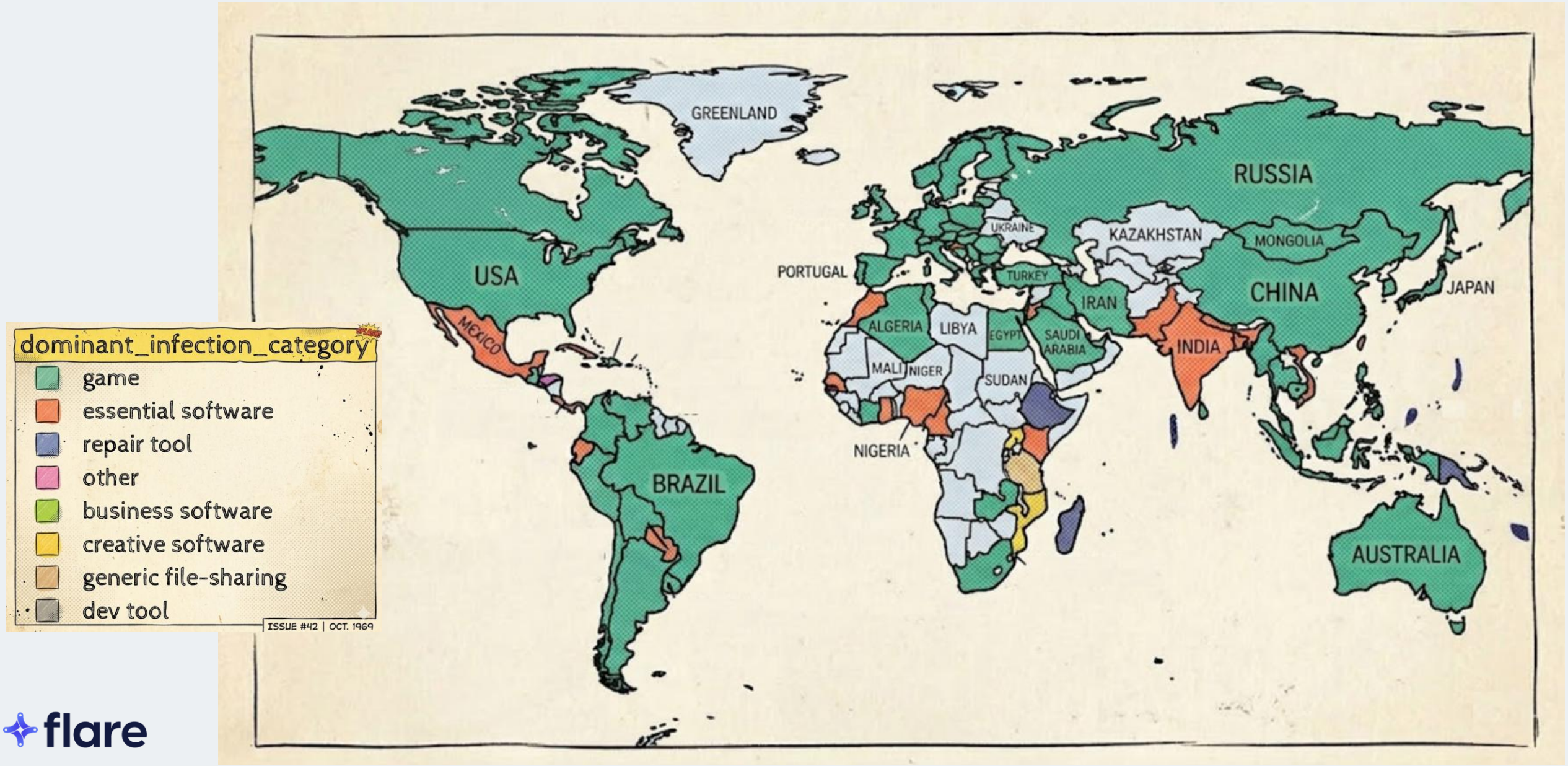
Examples:
Generic File download
AI
Repair Tools

Examples:
Adobe
Photoshop
Microsoft Office
AutoCAD



Examples:
Games (Minecraft, Roblox, Solitaire)
Porn

Dominant infection vector by country



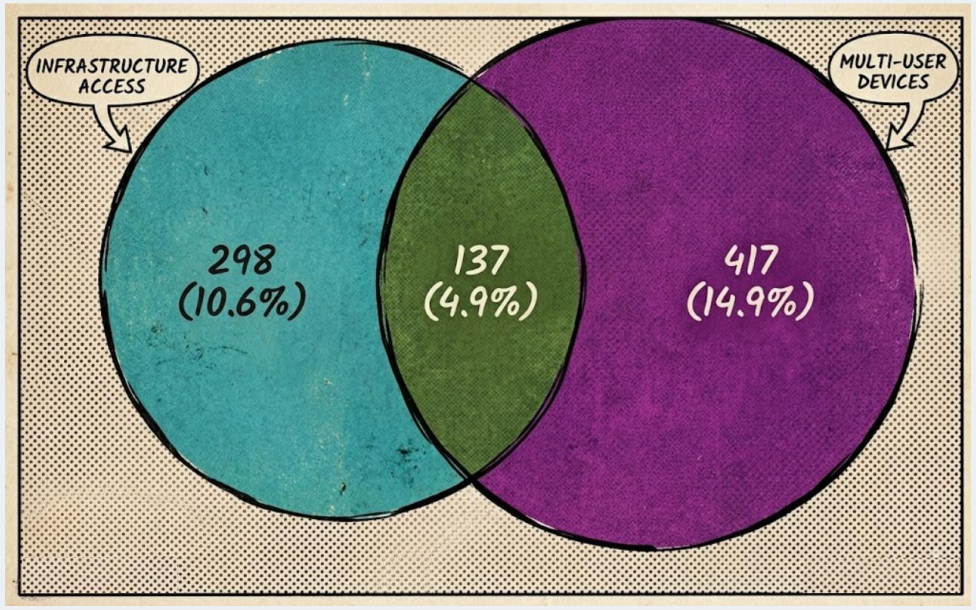
flare

Why you should care about game-related infections

16% of victims who have been infected by a game-related vector have company infrastructure access!

- Use the computer for personal use (15.5%)
- Share the corporate computer (5%)

Within Game Infections



26% of information stealer victims have access to company infrastructure

PERSONAL USE ON YOUR WORK COMPUTER CAN TOPPLE THE COMPANY.



FLARE-APPROVED WARNING

Scientifically verified through a year of daily infections from a dataset of more than 100M+ victims of stolen passwords.

Flare.io flare

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Victim Profiling

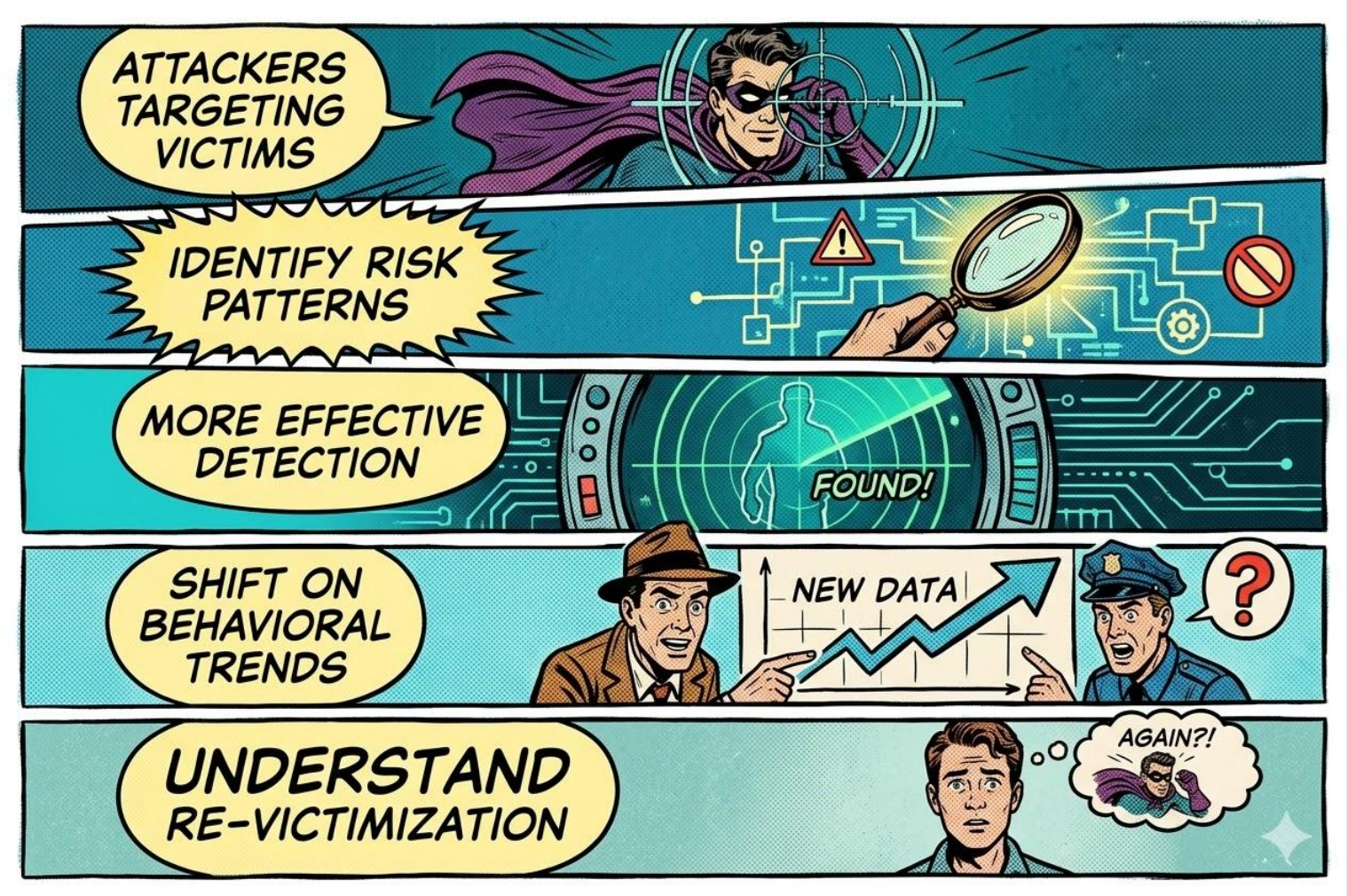
Identify to help prevent



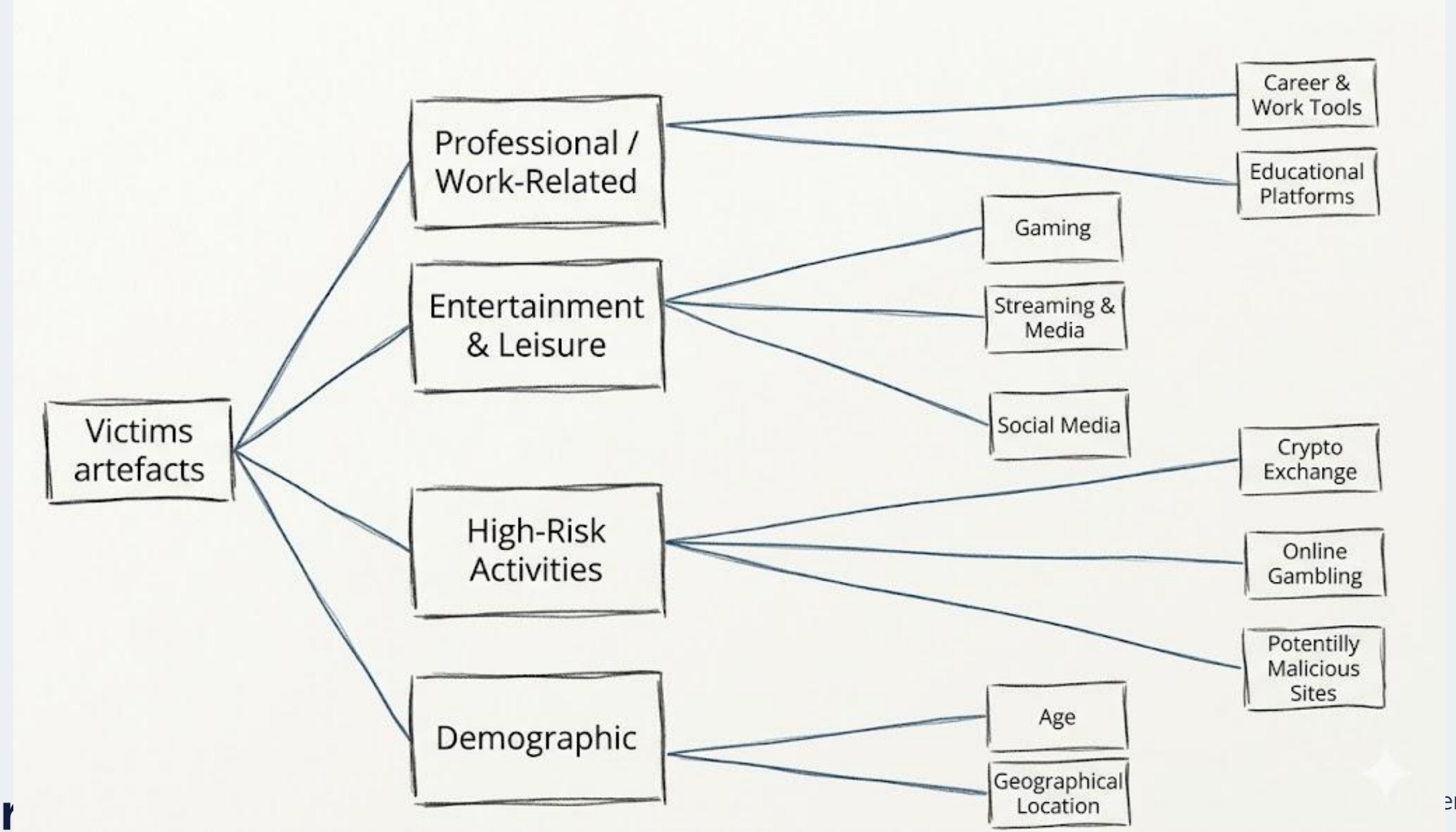
Victim Profiling

- Spot patterns, traits, and behaviors that make people more vulnerable
- Understand how context, habits, and demographics increase risk
- Analyze victim trends to improve prevention, awareness, and protection

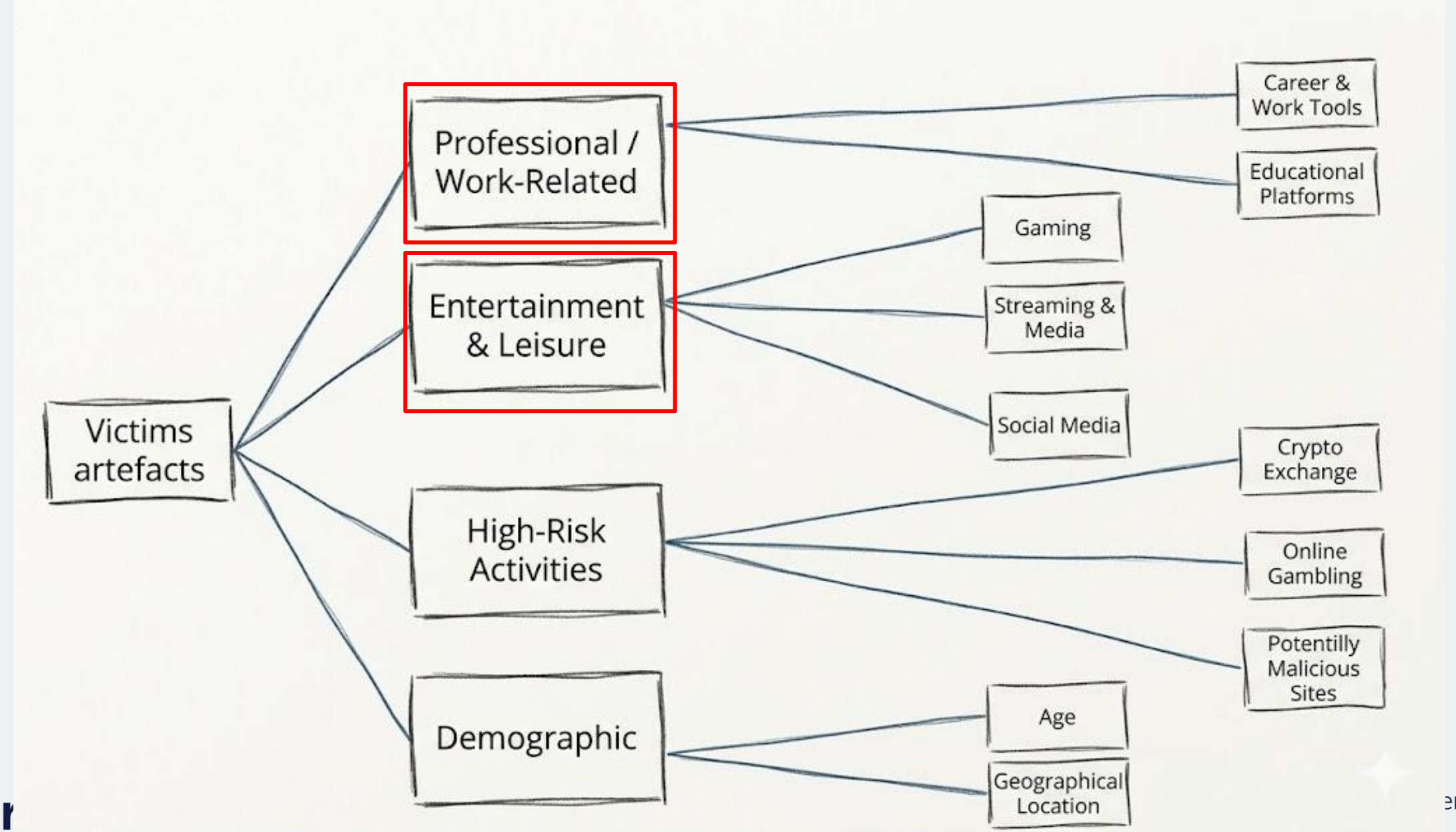
Victim Profiling in Cybersecurity



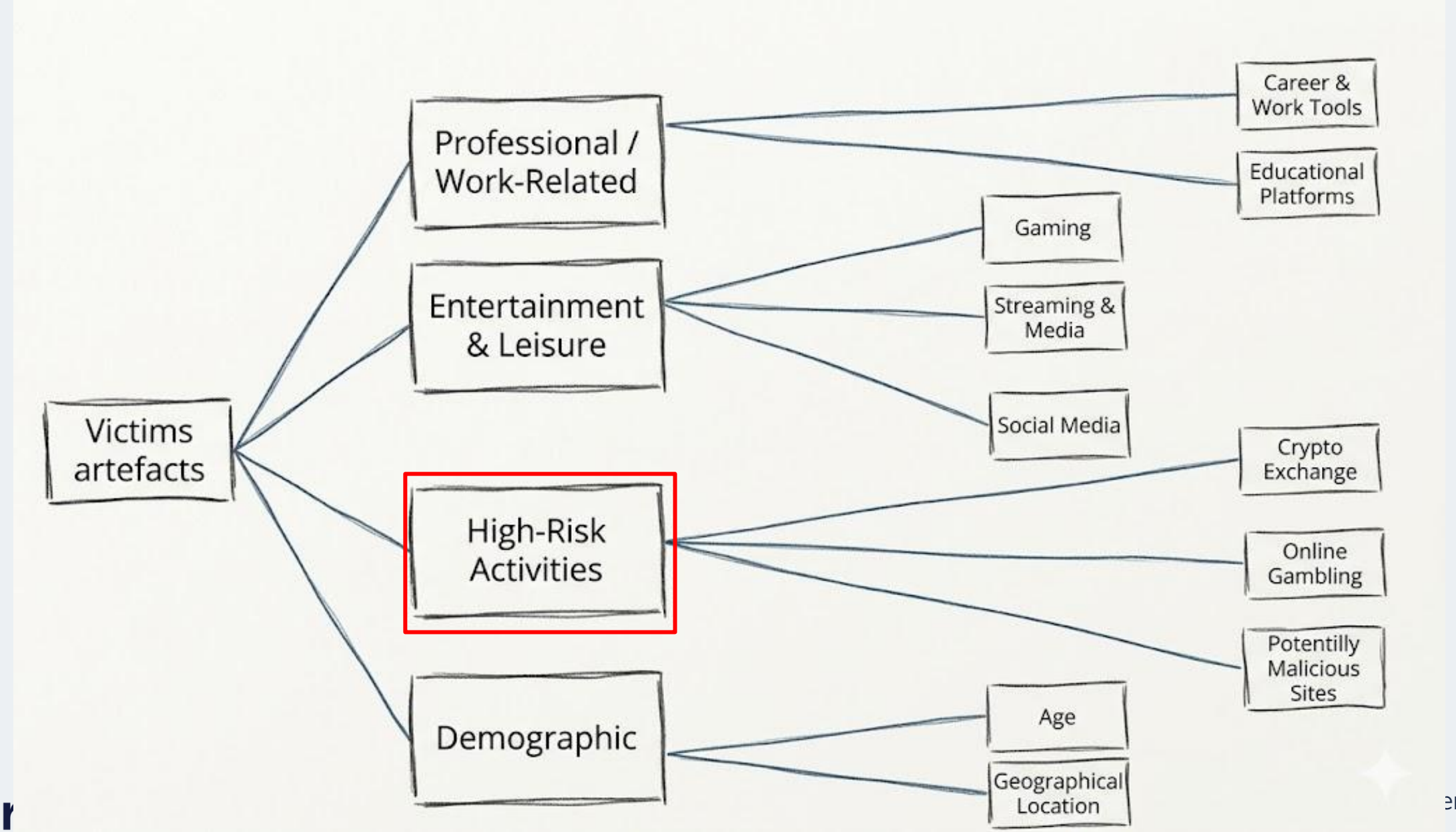
Variables Behind the Analysis



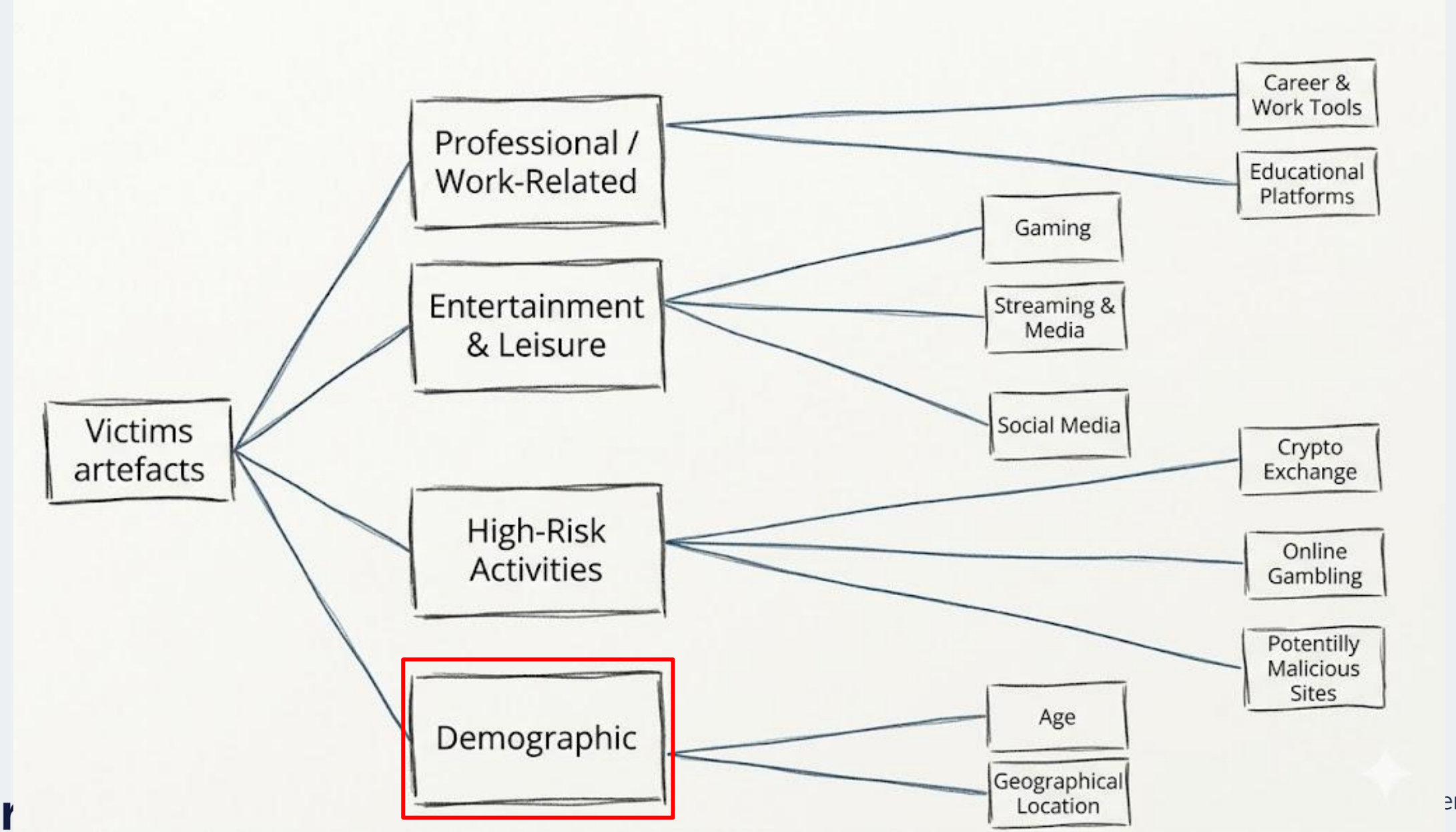
Variables Behind the Analysis



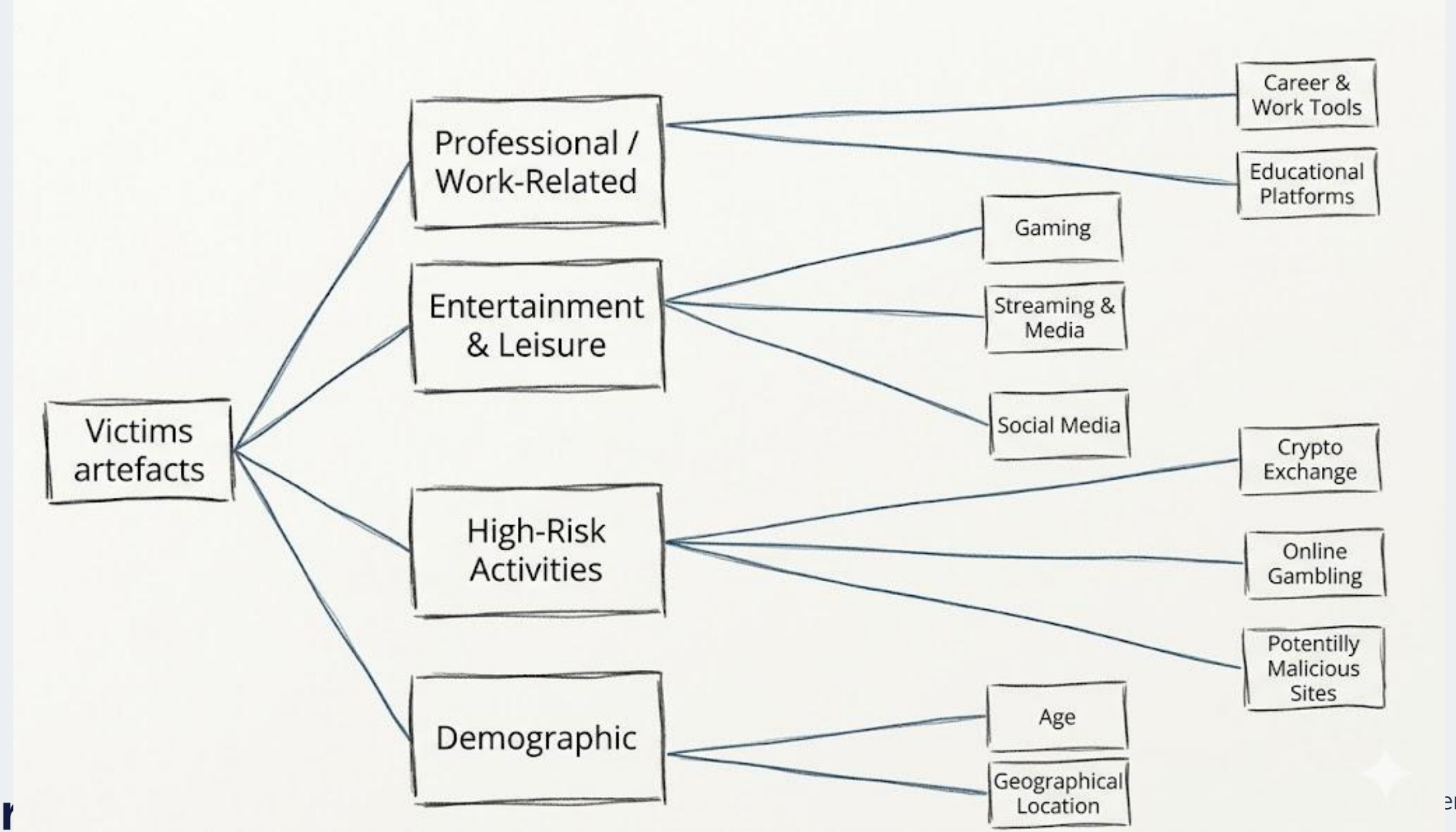
Variables Behind the Analysis

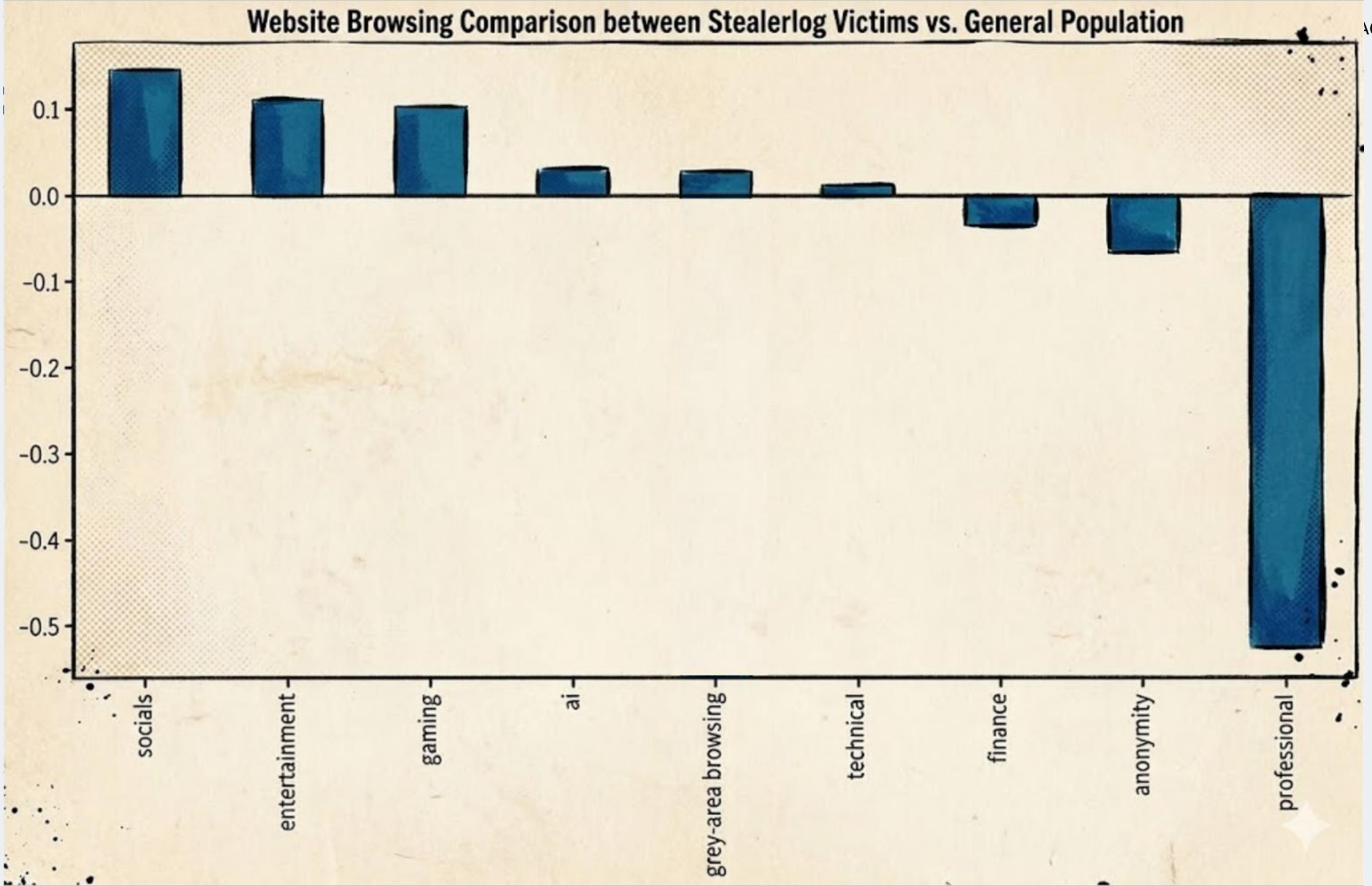


Variables Behind the Analysis



Variables Behind the Analysis




Website Browsing Comparison between Stealerlog Victims vs. General Population
0.1
0.0
-0.1
-0.2
-0.3
-0.4
-0.5
socials
entertainment
gaming
ai
grey-area browsing
technical
finance
anonymity
professional

Victims' browsing patterns

Overrepresentation of browsing in:

- social ecosystems
- entertainment ecosystems
- gaming ecosystems

Which are known to host:

- download portals
- game mods
- cracked software

2 Interacting Mechanisms

Cyber victimization may result from two interacting mechanisms:

1. **Structural exposure**: Users spend time in ecosystems where malware circulates.
2. **Behavioral predisposition**: Users engage in risk-taking or impulsive digital behaviors.

When both are present, victimization probability increases.

Important Numbers

Offending can increase the risk of later victimization by as much as

68%

in certain environments

- National Institute of Justice (NIJ)

Backdoored Threat Actor

Fortunately for defenders, bad guys are always one after another. There is a substantial amount of Threat Actors getting infected via tools that are backdoored.

Examples found:

- Compromised Italian Threat Actor via a FUD Crypter
  - Stealerlogs within the stealerlog
  - 287 credentials
  - History filled with hacking tools (IP Killer 2, Phoenix Botnet, etc.)
- North Korean IT Workers



Important Numbers

30%

of all breaches in 2025 were linked to a third-party

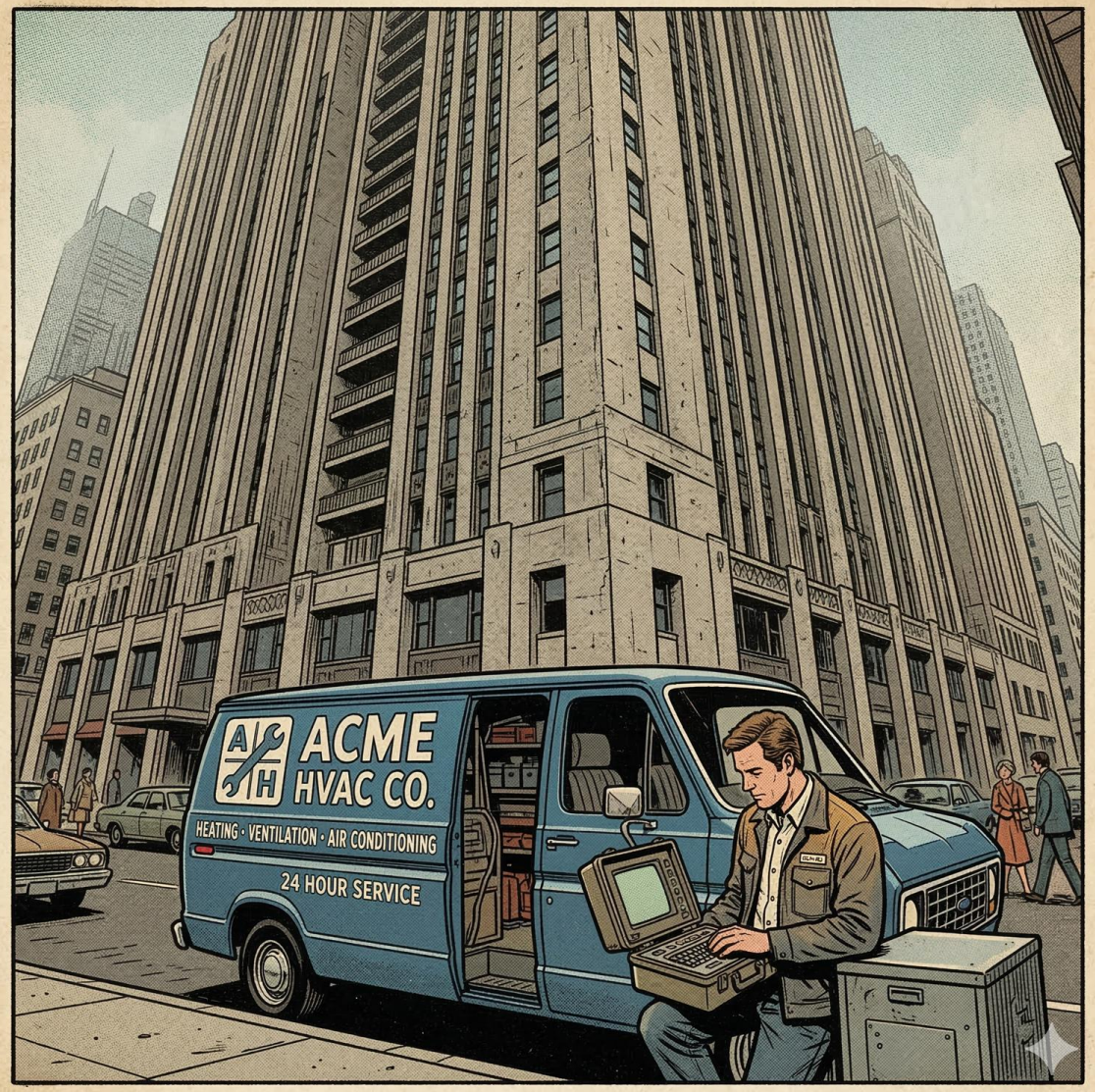
– Verizon DBIR 2025

Supply Chain Risks

3rd party access into your environments equals exposure. If the provider's security awareness is insufficient, it could lead to infection that exposes privileged access. This extends to software supply chain risks too.

Example found:

- A Nigeria-based digital agency
  - 3rd party cloud and SaaS access
  - cPanel, Zoho, HubSpot, etc.
  - Thousands of credentials, hundreds of emails+phones



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What Have We Learned?



The InfoStealer Malware Ecosystem

- is a significant threat
- is an opportunistic threat
- affects organizations via collateral effect
- prevention cannot be only technological, it is an awareness issue

Apply What You Have Learned Today

- Next week you should:
  - Distribute our StealerLog awareness posters digitally and on premises
  - Identify where critical credentials are stored in endpoints
    - What is outside SSO? Are Browser-based password managers disabled?
- In the first three months following this presentation you should:
  - Integrate educational material about StealerLogs in company cybersecurity awareness training
  - Promote password manager adoption

Apply What You Have Learned Today

- Within six months you should:
  - Cybersecurity educational material should be repeated and adapted
    - Cover new threat vectors, for example: Clickfix style
    - Adjust to the evolving audience, for example: more gaming risks if dealing with parents
  - Design in-house awareness posters based on real internal incident response cases
  - Proactively monitor for exposed credentials, session cookies and non-human identities (NHI)
    - Automate remediation where possible
  - Have all critical assets behind Single-Sign-On (SSO)

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Thank you!

Andréanne Bergeron
- Email: andreanne.bergeron@flare.io
- Social: https://www.linkedin.com/in/andreanne-bergeron-phd/

Olivier Bilodeau
- Email: olivier.bilodeau@flare.io
- Other hat: https://nsec.io
- Social: @obilodeau.bsky.social



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